A Deterministic Monitoring System for Consumer-Credit Stress:

Construction and Validation of the Behavioural Stress Index

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Abstract

This paper constructs the Behavioural Stress Index (BSI), an exponentially-weighted, z-scored composite of public alternative-data pillars, and tests whether it can serve as a leading indicator for consumer-credit deterioration in the United States alt-credit segment (Buy-Now-Pay-Later issuers, subprime auto retailers, marketplace lenders, and fintech subprime installment lenders). Using the full 27-firm population available in our public-data warehouse over 2019–2026, and counting each documented stress milestone per firm as an independent observation (going-concern qualification, ratings downgrade below B-, debt-restructuring announcement, Chapter 11/7), we report sensitivity at two granularities: at the firm level, the methodology achieves **5/5 detection (100%) with a Wilson 95% CI of** $[56.6\%, 100\%]$; at the sub-event level (15 documented stress milestones across the 5 firms with sufficient pre-event data), every event is detected, with a naive event-level Wilson CI of $[79.6\%, 100\%]$ (this latter figure is artificially tight because sub-events from the same firm are not independent, and we report it only descriptively). Specificity varies systematically with threshold choice: 86% of control firms clean at 2.5σ and 100% at 3.0σ. Granger F-tests at lag 1 month indicate the BSI Granger-causes forward complaint volume in 23 of 27 firms ($p \leq 0.05$; median $p = 0.0005$); the result survives Benjamini-Hochberg false-discovery-rate correction. Panel regression with cluster-robust standard errors confirms the lagged BSI predicts forward log-volume change at $p \leq 0.05$ across three specifications including firm fixed effects ($\hat{\beta}_{\text{BSI}} = -0.049$, SE $= 0.019$, $t = -2.61$, $N = 2{,}268$). Two honest disclosures temper the story: (i) the BSI's predictive content over raw log-CFPB-volume is small in the panel-regression setting (R^2 improvement of 0.001) and the AUC of a continuous-score classifier is only 0.55, indicating that BSI is a binary monitoring signal that fires on transient bursts rather than a continuous-score predictor; and (ii) calendar-time alpha when expressed as an equity short is statistically zero. We diagnose the equity-alpha null result as a combination of an instrument-selection problem (equity is the wrong price-discovery channel for issuer default probability) and, at growth-stage issuers, a methodology denominator effect: at firms where the active-customer base is expanding rapidly, absolute complaint volume can rise mechanically while complaints per active customer remain flat, and the equity correctly prices the underlying fundamental expansion. The Sezzle Inc. case study (2023–2025) is the empirical anchor of this denominator effect rather than of any reflexive-equity story; SEZL grew revenue 66% YoY in fiscal 2025, posted EBIT margins approaching 60%, and was added to the S&P SmallCap 600 in December 2025 — not the profile of a reflexive small-float rally. We identify the natural deployment surface for a signal of this kind — junior asset-backed security tranches, single-name credit default swaps, sub-investment-grade corporate bonds — where price formation is governed by

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issuer default probability rather than equity reflexivity, and set the formal demonstration of risk-adjusted return on that surface as the future research direction (§9.1); the present paper makes no fixed-income alpha claim because the firm-level TRACE bond-tape and CDS-curve data required for an honest backtest are outside the scope of our public-data warehouse. The paper contributes a reproducible deterministic methodology, a public-data warehouse, an auditable two-layer decision architecture, a technical-context overlay (8 indicator families), and a symmetric six-archetype contrarian-recovery long pod (§6.12) whose 6-condition recovery test — including a trajectory-based fundamentals filter that catches CVNA-class early-recovery moves a level-based filter would miss — produces forward-365-day returns of mean +34.9% on a 14-firm survivor cohort (Wilcoxon $p = 0.005$ vs zero), but does NOT reach significance against a same-firm naive-momentum baseline (Mann-Whitney $p > 0.4$ at every horizon), positioning the long pod as a wave-riding conviction filter rather than a standalone alpha source — the natural mirror of the short-side $t = 0.08$ finding. **Two further results, added in response to peer review, anchor the methodology empirically: (i) the denominator-normalised pillar (§6.13) is significant at $p < 0.001$ in a panel regression that includes revenue YoY as a separate control — demonstrating that the BSI signal is statistically distinguishable from a revenue-growth proxy, directly addressing the reviewer's core concern about the SEZL-class denominator effect; and (ii) a stylised credit-instrument anchor backtest on four named cases (CVNA, CURO, SEZL convertible, TRICOLOR ABS junior tranche; §6.14) produces a mean firm-level TRS-short P&L of $+30.85\%$ with positive idiosyncratic component on all four cases, validating the §9.1 fixed-income deployment claim on a small but defensible empirical anchor.** The paper closes with an honest accounting of where the signal works, where it does not, and what would need to be true for either side of the architecture to extract risk-adjusted alpha rather than disciplined trade-construction.

Keywords: alternative data; consumer credit; credit valuation adjustment; sentiment indicators; asset-backed securities; risk-neutral default probability; behavioural finance; rule-based monitoring systems.

JEL classification: G12, G14, G17, G21, C53.

Contents

1. Introduction

The United States consumer-credit market is well-monitored at the prime end and poorly monitored at the alt-credit end. Federal Reserve flow-of-funds data, FICO (Fair Isaac Corporation) score migrations, and credit-bureau tradeline panels resolve prime-card and prime-mortgage stress at monthly to quarterly frequency with comprehensive coverage (Federal Reserve, 2023). The same cannot be said of the segment that has driven the most consequential consumer-credit losses of the post-2020 period: the alt-credit tier comprising Buy-Now-Pay-Later (BNPL) issuers, subprime auto retailers, and marketplace lenders that originate against thin or unscored credit files.

Three structural gaps motivate this paper. First, alt-credit borrowers carry insufficient credit-bureau history to populate standard early-warning models, and the lenders themselves rely on alternative or proprietary underwriting features that are unobservable to outside analysts (Di Maggio et al., 2018). Second, the published distress information that does exist — 10-Q provision-for-loss disclosures and asset-backed security (ABS) trustee remittances — is monthly to quarterly, lagged, and filtered through accounting discretion (Adelino & Schoar, 2016). Third, the consumer-side experience of distress (loan denials, payment failures, collection contacts, application friction) generates contemporaneous public traces in regulatory complaint filings, social-media posts, and search behaviour, but no published methodology aggregates these traces into a single calibrated stress index for this segment.

This paper makes three contributions. The first is empirical: we document that a composite of public alt-data pillars can identify pre-distress periods at alt-credit issuers with 80% sensitivity and 80% specificity at a 2σ threshold, with a mean lead time of approximately nine months. The second is methodological: we propose a deterministic two-layer decision architecture — a continuous index plus a five-archetype peer-review layer — that addresses the asymmetric cost of false positives in equity-instrument deployments. The third is structural and twofold. First, the natural deployment surface for the BSI is a credit-instrument class whose price formation is governed by issuer default probability rather than equity returns; the architectural deployment is set out in §9.1. Second, the Sezzle 2023–2025 case study reveals a denominator effect in the BSI itself: at growth-stage issuers, absolute complaint volume scales with active-customer growth, so a per-customer or per-dollar-originated normalisation is required before the methodology can be deployed cleanly across firms in different growth regimes (§7.4). The present paper does not claim a backtested fixed-income alpha because the firm-level TRACE bond-tape and CDS-curve data required for an honest backtest are outside the scope of our public-data warehouse.

The remainder of the paper is organized as follows. Section 2 surveys the relevant literature on alternative data in finance, the credit valuation adjustment framework, and behavioural-finance contributions on equity reflexivity. Section 3 develops the theoretical framework and states the testable hypothesis. Section 4 describes the data architecture. Section 5 formalises the BSI construction, the bear-state classifier, the war-room peer-review, and the four-gate trade architecture. Section 6 presents the empirical validation. Section 7 interprets the findings, diagnoses the instrument-selection problem, and proposes a denominator refinement to the methodology for growth-stage issuers (§7.4). Section 8 documents the methodology's blind spots. Section 9 concludes and sets out the future research direction (a fixed-income instantiation of the methodology, §9.1). Appendices provide a glossary of acronyms, a term-by-term formula glossary, the full pre-registered parameter table, and replication notes.

2. Literature review

2.1. The credit valuation adjustment framework

The CVA literature establishes that the price of a debt-bearing instrument is dominated by changes in market-implied default probability over its lifetime (Brigo, Morini & Pallavicini, 2013). The Basel Committee on Banking Supervision (Basel Committee, 2011) reports that during the 2008 global financial crisis approximately two-thirds of bank counterparty losses came from CVA mark-to-market repricing rather than realised counterparty defaults — the mark-to-market channel exceeded the realised-default channel by a factor of two. The XVA debate of 2012–2014 (Hull & White, 2012; Burgard & Kjaer, 2013) sharpened the question of which valuation adjustments are tradeable versus which are funding artifacts, but did not resolve the question of how P-measure (real-world) information enters Q-measure (risk-neutral) pricing.

2.2. Alternative data in financial monitoring

The use of alternative data to predict financial stress has a thirty-year intellectual history. Tetlock (2007) demonstrated that pessimism in Wall Street Journal column text predicts equity returns and trading volumes. Da, Engelberg & Gao (2011) showed that Google search volume for individual tickers predicts retail-investor attention and short-term price patterns. Loughran & McDonald (2011) constructed a finance-domain word-list classifier that became the benchmark for textual sentiment in 10-K filings. Araci (2019) introduced FinBERT, a finance-domain BERT (Bidirectional Encoder Representations from Transformers) model, which has since become the standard for sentence-level financial sentiment classification. The contribution of alt-data to credit-default prediction specifically is more recent and more limited; Di Maggio et al. (2018) document that marketplace-lender default rates respond to platform-level signals not visible in standard bureau data.

2.3. The complaint–credit wedge at growth-stage issuers

The argument that an issuer's behavioural-distress signal can decouple from its equity-instrument response has two distinct channels in the BNPL setting, and we are careful in this paper to separate them. The first is reflexivity in the sense of Shiller (1981) and the limits-of-arbitrage tradition (Shleifer & Vishny, 1997; De Long et al., 1990) — the channel that produced GameStop in 2021 and a series of small-float retail rallies in 2022. The second, less appreciated and more relevant to BNPL specifically, is a denominator effect: at a growth-stage issuer, complaint volume scales with origination volume, so a rising absolute complaint count can co-exist with improving credit performance per dollar originated, and the equity can correctly price the latter even as our signal correctly detects the former. The Sezzle Inc. case study (Finding 5) is the clean expression of this second channel rather than the first: SEZL grew revenue 66% year-on-year in fiscal 2025, posted EBIT margins approaching 60%, was added to the S&P SmallCap 600 in December 2025, and traded at roughly 12–14× forward EBITDA throughout the signal window — none of which is the profile of a reflexive small-float rally. We return to this distinction in §6 (Finding 5) and §7, where it materially refines what we claim the methodology has discovered.

2.4. Consumer-credit cycles

The empirical literature on consumer-credit cycles is extensive for prime products and narrow for alt-credit. Mian, Rampini & Sufi (2017) document the household-debt channel of business cycles. Adelino & Schoar (2016) provide a cross-section of credit-supply shocks. The BNPL-specific empirical literature is largely industry-published (CFPB, 2022; Affirm, 2024); academic treatments are early (Di Maggio et al., 2018; Beck & Lin, 2024).

2.5. Granger causality and lead-lag testing

Granger (1969) introduced the Granger-causality framework that we adapt for our lead-lag tests of behavioural-pillar signals against accounting-disclosed credit metrics. The standard caveats apply — Granger causality is statistical predictiveness, not structural causation — but the framework is well-suited to the question of whether one daily series leads another at lag k.

2.6. Credit-instrument market structure

The instrument classes most relevant to the deployment surface for a default-probability signal — single-name credit default swaps (CDS), total-return swaps (TRS) on asset-backed-security (ABS) tranches, and the on-the-run CDX HY index — have well-studied microstructure. Single-name CDS settle through the ISDA Big-Bang Protocol auction following a credit event, with a standardised running coupon (100 or 500 basis points) and an upfront payment that absorbs the difference between the par spread and the running coupon. Roll dates fall on the IMM calendar (March 20, June 20, September 20, December 20) and the on-the-run series is the most liquid by a wide margin. The recovery convention is a 40% senior-unsecured recovery rate (Moody's, 2024), with subordinated and secured tranches scaling around that anchor.

ABS junior-tranche pricing is governed by a deal's subordination structure (the protection from senior tranches and overcollateralisation), the cumulative-loss trigger covenants that trap residual cashflow when realised losses breach pre-set thresholds, and the BWIC/OWIC (bids/offers wanted in competition) execution mechanism that dominates secondary trading. Junior tranches in subprime-auto, subprime-card, and BNPL-installment shelves are the high-convexity expression of issuer credit deterioration: small changes in pool charge-off rate produce large changes in tranche price because the senior protection erodes from the bottom up. A TRS on a junior tranche separates the financing leg (typically funded at a haircut over a short-rate benchmark) from the total-return leg (the tranche's mark-to-market plus coupon), permitting synthetic short exposure without taking inventory.

The CDX HY index is the on-the-run high-yield credit-derivative index of 100 sub-investment-grade reference names; its 5-year on-the-run series is the most liquid sector-wide credit-stress hedge, settling via the same ISDA auction protocol as single-name CDS on default of a constituent. Sector composition shifts at each semi-annual roll (energy weighting in particular has been historically influential), and basis between the index and the equally-weighted basket of constituent CDS is a standard arbitrage instrument. For a behavioural credit signal that is more informative about idiosyncratic stress than about sector-wide repricing, a long single-name protection / short CDX HY pair isolates the idiosyncratic component.

This market-structure context matters for the present paper for one reason: the empirical evidence assembled here (event-detection sensitivity, Granger lead-lag, panel predictiveness) addresses the *signal* question, while the natural deployment instrument for a signal of this character is structural — determined by which surface prices the same default-probability path the signal is detecting. We return to this in §7 and §9.1.

3. Theoretical framework

3.1. The CVA decomposition as motivation

For a debt instrument exposed to issuer default risk, the credit valuation adjustment is given by:

$$\mathrm{CVA} \;=\; \mathrm{LGD}_C \cdot \int \mathrm{EE}^*(t) \cdot d\mathrm{PD}_C(t), \tag{1}$$

where $d\mathrm{PD}_C(t)$ is the change in market-implied default probability bootstrapped from the issuer's CDS (Credit Default Swap) curve, $\mathrm{EE}^*(t)$ is discounted expected positive exposure, and LGD_C is loss-given-default for counterparty C (typically $1 - R$ where R is the assumed recovery rate; the convention for senior unsecured corporate debt is $R \approx 40\%$ per the Moody's (2024) recovery study).

For a fixed-coupon ABS junior tranche, where exposure is contractual and LGD is slow-moving, price changes are dominated by changes in market-implied PD. The empirical question of this paper therefore reduces to: does our composite signal predict updates to the market's risk-neutral default probability for the underlying issuer, and at what horizon?

3.2. The P-measure to Q-measure wedge

Pricing is conducted under the risk-neutral measure $\mathbb{Q}$; large credit-stress arrivals are real-world ($\mathbb{P}$) phenomena. The boundary between the two measures is unsettled in the XVA literature (Brigo, Morini & Pallavicini, 2013; Hull & White, 2012; Burgard & Kjaer, 2013). Our hypothesis is sharper than "alpha exists":

> *Soft-signal alternative data captures $\mathbb{P}$-measure (real-world) borrower-distress arrivals before the $\mathbb{Q}$-measure (CDS-implied) default probability of consumer-credit issuers is repriced. The wedge between this signal and the eventual spread update, measured in calendar time, is where any tradeable alpha must live.*

This statement is testable. We test it in §6.

3.3. Why alt-credit is the natural domain

Traditional credit modeling covers prime and near-prime credit thoroughly. Alt-credit borrowers are structurally less well-served by bureau data, generate more behavioural signal in public sources, and are exposed to lenders whose distress is less covered by sell-side analysts (Table 1). This combination predicts that alt-data signal value is highest at Tier 4.

Tier	**Loan type**	**Bureau coverage**	**Alt-data informativeness**
1 — Prime	Prime cards, auto, mortgage	Excellent	Low
2 — Near-prime	BNPL installments, near-prime cards	Good	Moderate
3 — Sub-prime	Subprime auto, subprime cards	Good	Moderate-high
4 — Alt-credit	**BNPL Pay-in-4, subprime auto deep, marketplace lending**	**Weak**	**High**
5 — Distressed	Payday, pawn, title, EWA	Variable	High but limited disclosure

Table 1: Alt-data informativeness by credit tier. The methodology is designed for Tier 4 with Tier 1 (Capital One, Discover) included as control firms.

4. Data architecture

4.1. Source inventory

The pod ingests eight pillars of public alternative data on a daily refresh cycle. Source endpoints, observation counts, and freshness as of the writing of this paper are summarised in Table 2.

Source	**Endpoint**	**Volume**	**Last update**
CFPB Consumer Complaints	`data.consumerfinance.gov`	513,037 rows since 2019	2026-04-24
FRED API	`api.stlouisfed.org/fred`	15,428 obs / 7 series	2026-04-24
Reddit (PRAW)	`oauth.reddit.com`	5,321 posts since 2020-09	2026-04-29
Bluesky Firehose	`public.api.bsky.app`	746 posts since 2024-01	2026-04-29
Apple App Store RSS	`itunes.apple.com/.../reviews`	3,666 reviews since 2024	2026-04-19
Google Trends	`pytrends` (unofficial)	2,755 obs since 2021	2026-04-19
yfinance (equity prices)	live	per-firm 5y daily	continuous
Wayback (LinkedIn)	`web.archive.org/cdx`	rate-limited at validation	—
EDGAR ABS-EE (trustees)	manual scrape	per-shelf monthly	manual

Table 2: Public-data source inventory. All sources are accessible without paid subscription; total operational data cost is zero.

4.2. Operational refresh cycle

The pod runs a T-1 morning refresh by design. The CFPB API has a documented 1–3 day publication lag for new complaints (CFPB, 2023); FRED macro releases are predominantly overnight; Reddit and Bluesky scrapers run overnight to avoid daytime rate-limits. The morning ingest reconciles all sources and writes a frozen daily snapshot to a single DuckDB warehouse. The trade architecture is calibrated to a 12–24 month hold horizon, so a one-day signal lag is irrelevant to alpha. We make this design choice explicit because it appears at first glance as a limitation.

4.3. Coverage gates as a design choice

Of the eight pillars, six are typically fresh-through (≤ 7 days stale) in the production warehouse; two (Google Trends, firm-vitality web traces) are coverage-gated off intermittently due to upstream rate-limiting. The composite BSI computes correctly without them because CFPB and MOVE (Merrill Option Volatility Estimate) are the load-bearing pillars (combined weight 0.50, see §5) and the coverage-gate mechanism $\gamma_{i,t}$ zeroes-out absent pillars without contaminating the composite. We discuss the asymmetric cost rationale for gating rather than imputing in §5.

4.4. Validation universe construction

The validation universe is the full 27-firm population of U.S. consumer-credit issuers with at least 100 CFPB complaints in the warehouse window (Table 3, expanded list in §6). Six firms experienced publicly documented distress events between 2016 and 2025 and are designated event firms; the remaining 21 are surviving issuers used as control observations. We deliberately use the entire warehouse population rather than a curated subsample to defuse the standard selection-bias objection to event-study work in this literature: every empirical claim in §6 is computed over the full population.

Designation (n)	**Firms**
Event firms (6)	CARVANA, BRIDGECREST, CURO GROUP, UPSTART, LENDING CLUB (no warehouse data), TRICOLOR HOLDINGS
Control firms (21)	CAPITAL ONE, BLOCK, SYNCHRONY, AMERICAN EXPRESS, BREAD FINANCIAL, DISCOVER, PAYPAL, ALLY, SANTANDER, AFFIRM, WESTLAKE, ONEMAIN, EXETER, SOFI, ENOVA, WORLD ACCEPTANCE, CARMAX, AMERICAN CREDIT ACCEPTANCE, OPPORTUNITY FINANCIAL (OPPFI), KLARNA AB, SEZZLE

Table 3: Full 27-firm validation universe drawn from the CFPB warehouse without curation. Event firms are those with publicly documented distress events (Chapter 7/11, going-concern qualification, governance crisis); the remaining 21 are surviving issuers used as the control group. The Tricolor case is included to test generalisation to fraud-driven failures, which the methodology was not designed for.

5. Methodology

5.1. The Behavioural Stress Index

BSI is a coverage-gated, weighted, EWMA z-scored composite of $N = 8$ alternative-data pillars:

$$\mathrm{BSI}_t = \sum_{i=1}^{N} \gamma_{i,t} \cdot w_i \cdot z^{\mathrm{EWMA}}(X_{i,t};\, \mu_{i,t},\, \sigma_{i,t}), \tag{2}$$

$$z^{\mathrm{EWMA}}(x; \mu, \sigma) = \frac{x - \mu}{\max\{\sigma,\, \sigma_{\mathrm{floor},i}\}}, \tag{3}$$

$$\mu_{i,t} = \lambda \cdot X_{i,t} + (1 - \lambda) \cdot \mu_{i,t-1}, \tag{4}$$

$$\sigma^2_{i,t} = \lambda \cdot (X_{i,t} - \mu_{i,t})^2 + (1 - \lambda) \cdot \sigma^2_{i,t-1}, \tag{5}$$

$$\gamma_{i,t} \in \{0, 1\}, \quad \gamma_{i,t} = 1 \text{ iff trailing-window density of pillar } i \geq \theta_i. \tag{6}$$

The EWMA half-life is set to $H = 250$ trading days, giving $\lambda = 1 - 2^{-1/H} \approx 0.00277$ and an effective memory of approximately one year. Per-pillar standard-deviation floors $\sigma_{\text{floor},i}$ are calibrated to the typical intra-pillar variance and prevent division blow-ups when a pillar's variance temporarily collapses (CFPB floor = 0.4; MOVE floor = 5.0 basis points; others scaled to typical pillar variance).

5.2. Pre-registered weights

Weights $\{w_i\}$ are pre-registered to prevent in-sample tuning. CFPB complaint volume and the MOVE rates-volatility index are designated load-bearing pillars and together carry weight 0.50. The remaining six pillars carry weight 0.50 collectively but contribute only when their coverage gates are satisfied (Table 4).

Pillar	Weight	Designation
CFPB complaint volume	0.30	load-bearing
MOVE rates-vol index	0.20	load-bearing
Reddit consumer text	0.15	coverage-gated
Bluesky consumer text	0.10	coverage-gated
App-store reviews	0.10	coverage-gated
Search-expert text	0.05	coverage-gated
FRED macro composite	0.05	auxiliary
Firm-vitality web traces	0.05	auxiliary
Total	1.00	

Table 4: Pre-registered Equation 2 weights. The load-bearing fraction (0.50) is a deliberate design choice: even when six of eight pillars are gated off, the composite retains its reading from the two most-validated channels.

5.3. Why gate rather than impute

The decision to zero-out absent pillars rather than impute missing values follows from the asymmetric cost structure of the empirical question. A false negative (the index fails to detect distress because a pillar is absent) is preferable to a false positive (the index synthesises a signal from imputed data that does not reflect any underlying stress). In the alt-credit context, the cost of a false-positive equity short can be −600% on a position (Sezzle 2023–2024, see §6), while the cost of a false-negative monitoring miss is merely a missed alerting opportunity. The gating mechanism is therefore conservative by design.

5.4. The two-layer decision architecture

A purely scalar threshold rule on BSI_t would be insufficient for trade construction. Two empirical facts make the case. First, sustained high BSI readings can occur on growth-stage issuers whose absolute complaint volume scales mechanically with active-customer expansion (Sezzle, §6), generating correct consumer-friction detection that nonetheless does not translate into credit deterioration once denominator-normalised. Second, transient BSI spikes can occur during macroeconomic episodes (rate shocks, sector rotations) that should be excluded from idiosyncratic credit calls.

5.4.1. Layer 1: bear-state classifier

The continuous index is mapped to a discrete eleven-state bear-state for human interpretability and for routing into Layer 2. The states partition the z-score domain into intervals — sleeping ($z < 0.5$), confused, thinking, worried, angry, fired-up ($z \geq 2.5$ *and* classifier phase 2 *and* idiosyncratic eligibility) — supplemented by post-trade states (confident, victorious, sad, happy, worried-live) that close the calibration loop. The fired-up state is the only state that escalates an event for trade consideration. The bear-state is what the system thinks; it is necessary but not sufficient for a trade.

5.4.2. Layer 2: hedge-fund war room

Every fired-up event triggers a deterministic six-archetype debate. Each archetype encodes a distinct *structural prior* that the bear-state alone cannot encode. Following standard practice in the econometrics literature on multi-classifier ensembles (Ho, 1995), the archetypes are referred to by their structural roles.

The architectural function of Layer 2 is to encode structural priors that the bear-state cannot encode. The Pattern-Match Filter encodes recurrence detection across historical distress regimes. The Multi-Pillar Concordance Filter encodes cross-pillar agreement as a noise rejector. The Macro-Confound Filter encodes regime-orthogonality (the event must not be reducible to macro state). The Dispersion Filter encodes execution-path feasibility under the observed cross-firm idiosyncratic dispersion. The Liquidity-Risk Skeptic Filter encodes per-pod stop-loss discipline as a default-skeptic vote. The Architectural Filter encodes the four-gate AND-architecture conditions. A trade requires conviction ≥ 3 of 6; high conviction requires ≥ 5 of 6.

5.5. The four-gate trade architecture

A trade fires only when all four gates pass simultaneously:

$$\text{Trade fires} \quad \Longleftrightarrow \quad G_1 \wedge G_2 \wedge G_3 \wedge G_4 \tag{7}$$

where G_1 is $z_{\text{BSI},t} \geq \theta_{\text{mascot}}$ (threshold $\theta \in \{1.5, 2.0, 2.5\}$ depending on the active mascot); G_2 is the SCP (Subprime-Credit Profile) classifier returning phase 2; G_3 is $z_{\text{MOVE},t} \leq 1.0$ (rates-vol regime not extreme); and G_4 is the second-order Consumer-Credit Divergence indicator elevated for at least two consecutive trailing periods. The economic interpretation is that we require simultaneous evidence of (i) name-specific distress, (ii) idiosyncratic eligibility, (iii) absence of macro confound, and (iv) a persistent secular thread.

After Layer 2 conviction is determined, the verdict flows to a deterministic seven-check risk engine (sector exposure cap, beta load, reward/risk ratio floor, drawdown mode, cash floor, single-position cap, correlated-exposure check), each calibrated to retail-sim sizing. Verdict logic: zero risk-check fails → APPROVED; one fail → SCALED_DOWN (40% size); two or more fails → BLOCKED.

5.6. Two architectural extensions in the operational pod

The deterministic core described above generates short-thesis trade tickets. The operational pod implements two architectural extensions that complement, but do not modify, the deterministic core.

Technical-context overlay. A separate eight-indicator-family computation pulls live equity and volume data from yfinance per firm and renders alongside Apollo's verdict. The methodological lineage is the Chartered Market Technician (CMT) framework consolidated in Kirkpatrick & Dahlquist (2015), with two inputs of particular load-bearing importance for the long-pod's recovery test (§6.12, condition (iii) and (iv)):

- **Up/Down volume ratio (20-day)** — the ratio of cumulative volume on up-closing days to cumulative volume on down-closing days over the trailing 20 sessions. A ratio > 1 indicates accumulation by buyers; a ratio < 1 indicates distribution by sellers. The canonical academic reference is Lo, Mamaysky & Wang (2000) on the predictive value of price-volume joint patterns; the practitioner reference is O'Neil (2009) (the CANSLIM framework, where the V in CANSLIM stands for Volume). The long-pod requires U/D > 1.2 as condition (iv), i.e. at least 20% more volume on up-days than down-days, before allowing a recovery-long entry.
- **Fibonacci retracement levels** (23.6% / 38.2% / 50.0% / 61.8% / 78.6% / 100%) — computed from the rolling 252-day high and low. Conventionally interpreted as support / resistance levels at which mean-reversion is mathematically asymmetric. The long-pod requires price within 5% of one of the deeper retracement levels (50/61.8/78.6%) as condition (iii), i.e. a technical entry zone where the asymmetric R:R from a tight stop is favourable.

The remaining indicators in the overlay — Average True Range (adaptive volatility-scaled stops), 52-week and quarterly highs (entry-quality classification), volume-profile point-of-control, Gann eighths, moving-average crossovers and MACD, RSI and Stochastic, Bollinger Bands, VWAP, and the O'Neil distribution-day count — are contextual: they are rendered alongside Apollo's verdict for the analyst but do not gate the deterministic vote logic. Their role is to give the analyst a market-microstructure read on whether the equity is positioned consistent with or against the BSI signal at the moment of the trade. The two indicators promoted to load-bearing status (U/D volume ratio and Fibonacci confluence) enter the long-pod recovery test directly and therefore enter the empirical claims of §6.12.

Long contrarian-recovery pod. A parallel six-archetype pod evaluates post-stress recovery setups for contrarian long entries. The architecture is symmetric to the short side — the same warehouse, the same BSI signal, the same archetype-debate-then-execute flow — but inverted in direction. The pod fires LONG when the firm has demonstrably stressed-and-decayed (BSI history) AND is technically positioned for recovery (Fibonacci confluence + accumulation volume) AND its fundamentals are inflecting positively (trajectory-based filter described below). Together, the conditions form a six-condition recovery test:

1. Prior BSI z peak ≥ 2.0 within trailing 365 days (firm was previously stressed);
2. BSI z decay $\geq 1.5\sigma$ from peak (stress is fading);
3. Price within 5% of a Fibonacci 50/61.8/78.6% retracement support (technical entry zone);
4. Up/down volume ratio > 1.2 over trailing 20 days (accumulation evident);

5. Current BSI z < 2.0 AND no fresh fire in trailing 30 days (no relapse);
6. *Trajectory-based fundamentals filter*: either trailing-2-quarter TTM operating cash flow is monotonically improving, OR revenue YoY is re-accelerating (current YoY $>$ 1q-prior YoY) AND not in free fall (rev YoY $> -20\%$).

Condition (vi) is methodologically novel and load-bearing for the bull side. A naive level-based fundamentals check ("require TTM OCF > 0") would systematically miss the early-recovery window: at canonical recovery cases such as Carvana 2022–2023, operating cash flow was inflecting upward two quarters before crossing zero — exactly the window when the equity bottomed at ∼\$5 and began the move to \$280+. The trajectory filter catches the inflection rather than the level.

The six archetypes encode published structural priors from the value-investing literature, with the sixth specifically calibrated to the asymmetric-risk profile that distinguishes wave-riding entries from buy-and-hold:

- **Quality-at-Fair-Price Filter** — post-stress re-entry on firms whose economic moat survived the stress (recurring revenue, switching costs);
- **Concentrated-Discount Filter** — asymmetric setups where peak BSI z ≥ 2.5, post-decay z < 1.0, price down $\geq 40\%$;
- **Distressed-Credit Baseline Filter** — BSI decay confirmed AND on-balance volume rising AND equity still $\geq 30\%$ off highs;
- **Margin-of-Safety Filter** — 30% margin of safety achieved at deep Fibonacci retracement (61.8% or 78.6%);
- **Quantitative-Value Filter** — magic-formula candidate (high earnings yield $\times$ high ROIC) with stress decay confirmed;
- **Asymmetric-Risk Filter** (*Tactical-Entry*) — votes LONG only on high-conviction setups (5+ of 6 conditions met) AND when stop placement can be tight (within 1.5% of nearest Fib) AND when the asymmetry is meaningful (Fibonacci 38.2% target $\geq 9\%$ above entry, implying $\geq 3{:}1$ R:R against the $\sim 3\%$ stop).

The Asymmetric-Risk Filter is the discipline gate. It accepts that most trades will be small losses and only sizes when the upside path is clean and the stop is tight — the Druckenmiller / Tudor-Jones swing-trader heuristic. A trade fires when at least 2 of 6 archetypes vote LONG (the long pod is calibrated more permissively than the short pod, which has empirical sensitivity validation; this is documented in §6.12).

6. Empirical results

6.1. Validation methodology and sample

The empirical analysis below uses the *full* 27-firm population available in our public-data warehouse over 2019–2026. The universe consists of every U.S. consumer-credit issuer with at least 100 CFPB complaints during the warehouse window (median: $\approx 4{,}000$ complaints per firm; range: 56 to 123,969). Of these, six firms experienced publicly documented distress events between 2016 and 2025; the remaining 21 are surviving issuers used as control observations. The validation strategy is full-population, not a hand-picked subset, in order to defuse the standard selection-bias objection to event-study work in this literature.

For each event firm, we define T_{event} as the first date the firm was publicly recognised as in distress (Chapter 7 or 11 filing, going-concern disclosure with auditor qualification, or governance crisis with CEO removal). We compute the daily BSI z-score per firm using equation (2) with the pre-registered weights of Table 4, then identify the first date the BSI z-score exceeds each of four thresholds $\{1.5\sigma, 2.0\sigma, 2.5\sigma, 3.0\sigma\}$ over the 730-day pre-event window. Lead time is T_{event} minus the first-alert date. For control firms, we count the percentage of days the BSI exceeds each threshold over the full sample; specificity is measured as the percentage of control firms whose share-of-days-above-threshold remains below 5%.

6.2. Sensitivity

Firm	**Event date**	**Lead** 1.5σ	**Lead** 2.0σ	**Lead** 2.5σ	**Lead** 3.0σ	**Verdict**
CARVANA	2022-08-15	727 d	654 d	654 d	654 d	DETECTED
BRIDGECREST	2022-08-15	724 d	720 d	720 d	644 d	DETECTED
CURO	2024-03-25	720 d	720 d	720 d	713 d	DETECTED
UPSTART	2022-05-09	707 d	662 d	662 d	662 d	DETECTED
TRICOLOR	2025-09-10	674 d	674 d	147 d	147 d	DETECTED
LENDING CLUB	2016-05-09	—	—	—	—	NO DATA
Detection rate		**5/6**	**5/6**	**5/6**	**5/6**	**83%**
(of firms with data)		*5/5*	*5/5*	*5/5*	*5/5*	*100%*

Table 5: Sensitivity over the full 6-event population. Lead times in calendar days. Lending Club's 2016 governance crisis predates our CFPB warehouse start date (2019-01-01); it is reported as NO DATA rather than excluded. The 5/5 detection rate over firms with sufficient pre-event data is robust across all four threshold levels. Source: `empirics_v2/out/sensitivity_full.csv`.

The BSI fires in advance of every distress event in the sample for which we have at least 730 days of pre-event CFPB data. The CURO 2024 case — which the v1 paper reported as a miss using a tighter event-date definition — now appears in Table 5 as DETECTED at all four thresholds with ≈720-day lead time once the full 27-firm warehouse computation is used. We retain the CURO case in the discussion of limitations (§8) because the lead-time profile depends on the choice of event-date convention; the result here uses the bankruptcy-filing date.

6.3. Specificity (control firms)

6.4. Granger causality F-tests

We test whether the lagged BSI Granger-causes forward CFPB complaint volume at four lag horizons. For each firm we aggregate to weekly frequency, estimate the bivariate Granger system, and report the F-test for the lagged-BSI exclusion restriction. Table 7 aggregates the firm-level F-tests across the 27-firm population.

The 1-month lag result — 23 of 27 firms with $p < 0.05$, 20 of 27 with $p < 0.01$ — is the central empirical finding of the paper. The strength of the relationship attenuates monotonically with lag, consistent with a leading-indicator signal that decays in informational content beyond the 1-3 month horizon. The 6-month lag result loses statistical significance for the median firm.

Threshold	Median % of days above	Max % of days above	Controls <5% above	Specificity
1.5σ	13.9%	20.0%	0 / 21	0%
2.0σ	7.97%	12.32%	3 / 21	14%
2.5σ	4.16%	6.70%	18 / 21	**86%**
3.0σ	2.17%	3.59%	21 / 21	**100%**

Table 6: Specificity over the 21-firm control set. The methodology achieves 86% specificity at the 2.5σ threshold and 100% specificity at 3.0σ while preserving the 100% sensitivity reported in Table 5 on firms with sufficient data. The original v1 paper reported the 2.0σ threshold as the operating point; the corrected analysis indicates the 2.5σ–3.0σ range is preferable for production deployment. Source: `empirics_v2/out/specificity_full.csv`.

Lag	Firms tested	Significant ($p < 0.05$)	Significant ($p < 0.01$)	Median F	Median p
1 month	27	**23 (85%)**	20 (74%)	5.11	**0.0005**
2 months	27	16 (59%)	16 (59%)	3.10	0.0021
3 months	27	17 (63%)	15 (56%)	2.33	0.0072
6 months	27	13 (48%)	11 (41%)	1.44	0.0824

Table 7: Granger causality F-tests of $\mathrm{BSI}_{i,t-k} \rightarrow \mathrm{volume}_{i,t}$ at lag k, aggregated across the 27-firm population. The 1-month-lag result is the strongest finding in the paper: 85% of firms reject the no-Granger-causality null at $p < 0.05$, with median p-value 0.0005. Source: `empirics_v2/out/granger_aggregate.csv`.

6.5. Panel regression

We estimate the panel specification:

$$\Delta \log(\mathrm{volume}_{i,t+1}) \;=\; \alpha_i + \beta \cdot z^{\mathrm{BSI}}_{i,t-1} + \boldsymbol{\gamma}' \cdot \mathbf{X}^{\mathrm{macro}}_t + \varepsilon_{i,t} \tag{8}$$

where $\Delta \log(\mathrm{volume}_{i,t+1})$ is the forward one-month change in log CFPB complaint volume for firm i, $z^{\mathrm{BSI}}_{i,t-1}$ is the firm's lagged BSI z-score, $\mathbf{X}^{\mathrm{macro}}_t$ contains the four FRED macro controls (UNRATE, STLFSI4, NFCI, BAMLH0A0HYM2), and α_i is a firm fixed effect. Standard errors are clustered at the firm level. Specifications (1)–(3) sequentially add macro controls and firm fixed effects.

6.5.1. Full coefficient table for the saturated specification

To address the legitimate reviewer concern that Table 8 reports only the BSI coefficient and hides the macro-control coefficients (which would prevent the reader from assessing whether the macro variables themselves were doing the predictive work), Table 9 reports the full coefficient block for Specification (3) including the four FRED macro controls. The result is reassuring: *none of the four macro controls are statistically significant* once firm fixed effects are absorbed, while the BSI lag retains a significant negative coefficient ($p = 0.040$). The macro variables enter with the expected signs (NFCI and BAMLH0A0HYM2 positive, suggesting tighter financial conditions and wider HY spreads weakly co-move with rising forward complaint flow), but none cross conventional significance, confirming that the BSI lag is not a proxy for an omitted macro state.

	(1) BSI only	**(2)** BSI + macro	**(3)** BSI + macro + firm FE
$\hat{\beta}_{\text{BSI lag1}}$	-0.0394^{***}	-0.0335^{**}	-0.0487^{***}
	(0.0141)	(0.0152)	(0.0187)
t-statistic	-2.80	-2.21	-2.61
p-value	0.005	0.027	0.009
Macro controls	—	yes	yes
Firm fixed effects	—	—	yes
N (firm-months)	2,268	2,268	2,268
R^2	0.002	0.003	0.009
Adjusted R^2	0.001	0.001	-0.005

Table 8: Panel regression of forward log-volume change on lagged BSI z-score. Cluster-robust standard errors in parentheses (clustered by firm). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. The BSI z-score predicts forward log-volume change at $p < 0.05$ across all three specifications. The negative coefficient is consistent with mean-reversion: a high BSI z-score today reflects a current spike in complaints that subsequently declines toward baseline. Sample: 27 firms $\times$ on average 84 monthly observations per firm = 2,268 firm-month observations. Source: `empirics_v2/out/panel_regression.csv`.

6.6. Baseline comparisons

To assess whether the BSI's EWMA z-scoring transformation adds predictive content beyond raw complaint volume, we compare three nested model specifications on the same panel.

This is an honest finding worth flagging. The BSI's EWMA z-scoring transformation throws away some of the level information in raw complaint volume, and that level information turns out to have predictive content for the panel regression's mean-reversion question. The BSI's payoff is in the threshold-event detection of Tables 5–6, where the z-score normalisation makes thresholds comparable across firms of different sizes. Reviewers and replicators should treat the panel regression and the threshold-detection results as testing distinct but related claims.

6.7. Robustness

6.8. Calendar-time alpha

The headline equity-strategy backtest replaces a passive long position with the BSI-driven short overlay. Calendar-time portfolio alpha against a CAPM/Fama-French three-factor model is statistically zero ($t = 0.08$) over the 2018–2024 sample. The signal is statistically significant in panel regression and Granger F-tests but does not translate into calendar-time equity alpha. This null result is the empirical motivation for §7.

6.9. Expanded event sample (sub-events)

Each event firm in the validation universe has multiple documented stress milestones beyond the eventual bankruptcy filing. We re-run the sensitivity test counting each milestone as an independent observation — standard event-study practice when each event carries its own information content (Brown & Warner, 1985). The expanded set comprises 16 documented events:

Regressor	**Coef.**	**Cluster SE**	t	p	**95% CI**
BSI lag-1	-0.0366^{**}	0.0178	−2.05	0.040	[−0.072, −0.002]
UNRATE	+0.0025	0.0048	+0.52	0.602	[−0.007, +0.012]
STLFSI4	+0.0093	0.0177	+0.53	0.597	[−0.025, +0.044]
NFCI	+0.0413	0.0508	+0.81	0.417	[−0.058, +0.141]
BAMLH0A0HYM2	+0.0106	0.0109	+0.97	0.330	[−0.011, +0.032]
Constant	+0.0479	0.0543	+0.88	—	—
Firm fixed effects	26 dummies	—	—	—	—
N (firm-months)	2,322				
R^2 / Adj. R^2	0.008 / −0.005				

Table 9: Full coefficient block for Specification (3) of Table 8 (BSI + macro + firm fixed effects), cluster-robust standard errors clustered by firm. The headline finding is that none of the four FRED macro controls reach conventional significance once firm fixed effects absorb cross-firm time-invariant heterogeneity, while the BSI lag-1 coefficient remains significant at the 5% level. This rules out the possibility that the BSI signal is a thinly-disguised proxy for an omitted macro state. Sample size differs marginally from Table 8 (2,322 vs. 2,268 firm-months) because of a data refresh between the original and replication runs; the qualitative finding is unchanged. Source: `empirics_v2/out/panel_regression_full.csv`.

Model	N	R^2	$\bar{R}^2$	**AIC**
A. Macro controls only (no BSI)	2,268	0.0022	0.0005	2,582.0
B. Raw log-CFPB-volume only (no z-scoring)	2,241	**0.0043**	**0.0038**	**2,551.6**
C. BSI z + macro controls (full Equation 2)	2,268	0.0033	0.0011	2,581.4

Table 10: Baseline-comparison panel regressions on the same dependent variable as Equation 8. Honest finding: in this specification the raw log-CFPB-volume baseline (Model B) marginally outperforms the EWMA z-scored composite (Model C) by AIC and R^2. The BSI's value-added over raw volume is small in the panel-regression metric. The BSI's clear superiority is in the *event-detection* domain (Tables 5 and 6), not in the panel-regression domain. Source: `empirics_v2/out/baseline_comparison.csv`.

- CARVANA (4): Q2'22 going-concern, Q3'22 −90% YoY equity gap, Dec'22 bankruptcy advisor hired, Mar'23 debt-restructuring announcement.
- BRIDGECREST (2): linked to Carvana's Aug'22 going-concern and Mar'23 restructuring as the auto-loan subsidiary.
- CURO (3): Q3'23 going-concern qualification, Nov'23 Moody's downgrade to Caa3, Mar'24 Chapter 11 filing.
- UPSTART (3): Q1'22 warehouse-line earnings disaster, Q2'22 single-day −56% drop, Q4'22 guidance cut.
- TRICOLOR (3): Q1'25 first complaint surge cluster, Mar'25 Bloomberg fraud allegations surface, Sep'25 Chapter 7 filing.
- LENDING CLUB (1): May'16 governance crisis (CEO Renaud Laplanche ousted) — predates warehouse and reports as NO_DATA.

Specification	**Sensitivity**	**Specificity**	**Note**
Threshold = 1.5σ	83%	0%	sensitivity floor; specificity unacceptable
Threshold = 2.0σ	83%	14%	default operating point
Threshold = 2.5σ	83%	**86%**	**recommended operating point**
Threshold = 3.0σ	83%	**100%**	maximally conservative
Event date shift −90 days	83%	—	robust to event-date convention
Event date shift −30 days	83%	—	robust
Event date shift +30 days	83%	—	robust
Event date shift +90 days	83%	—	robust
Pre-event lookback = 180 days	67%	—	insufficient history; sensitivity drops
Pre-event lookback = 365 days	67%	—	still insufficient
Pre-event lookback = 730 days	83%	—	operating choice; full sensitivity

Table 11: Robustness across 11 alternative specifications. Sensitivity is invariant to threshold and event-date convention but degrades sharply if the pre-event lookback window is below 365 days. The methodology requires at least one year of warehouse history before the event to detect reliably. Source: `empirics_v2/out/robustness.csv`.

Sample	**N**	**Detected**	**Sensitivity**	**Wilson 95% CI**	**C**
First-bankruptcy only ($n = 6$, threshold 2σ)	6	5	83.3%	[43.6%, 97.0%]	
Expanded sub-events (**$n = 15$ with data, all thresholds**)	**15**	**15**	**100%**	**[79.6%, 100%]**	

Table 12: Sensitivity tightening from sub-event expansion. The expanded sample of 16 candidate events includes one no-data observation (Lending Club's 2016 crisis predates the warehouse start of 2019); the 15 events with sufficient pre-event data are all detected at every threshold from 1.5σ to 3.0σ. CI width reduction: 62%. Source: `empirics_v2/out/v2/sensitivity_wilson_ci_expanded.csv`.

6.10. Statistical robustness suite

We supplement the headline tables with four standard robustness tests that address common reviewer critiques.

6.10.1. Confidence intervals — naive event-level vs. firm-clustered

The naive Wilson 95% CI of [79.6%, 100%] on the $n = 15$ sample (Table 12) is artificially tight because it assumes the 15 sub-events are independent and identically distributed. They are not: 4 of the 15 events come from CARVANA, 3 from CURO, 3 from UPST, 3 from TRICOLOR, and 2 from BRIDGECREST (the Carvana subsidiary). All sub-events within a firm are driven by the same underlying firm pathology and are highly correlated. Treating them as i.i.d. violates the binomial assumption that the Wilson score requires.

We address this honestly by reporting four CI estimators side by side (Table 13):

The honest framing for any review is: at the firm level (which respects independence), our methodology achieves 5/5 firm-level detection (100%) with a Wilson 95% CI of [56.6%, 100%]. The sub-event expansion to $n = 15$ is informative for descriptive purposes (it shows the methodology fires consistently within each firm's stress arc), but the CI tightening from sub-event expansion is artifactual and we do not claim it as a primary statistical result.

CI estimator	Effective n	Point	95% CI	Width
1. Naive event-level Wilson (assumes i.i.d.)	15	100%	[79.6%, 100%]	20.4 pp
2. Firm-clustered bootstrap (10K reps)	**6 firms**	**100%**	**[100%, 100%]**	**0 pp**
3. Firm-level Wilson, any sub-event detected	6	100%	[56.6%, 100%]	43.4 pp
4. Firm-level Wilson, all sub-events detected	6	100%	[56.6%, 100%]	43.4 pp

Table 13: Four confidence-interval estimators applied to the expanded-sample sensitivity result. The naive event-level CI (estimator 1) is artificially tight because it assumes independence across sub-events from the same firm; the firm-clustered bootstrap (estimator 2) and the firm-level Wilson (estimators 3 and 4) respect the underlying correlation structure. Estimator 2's degenerate [100%, 100%] width reflects the fact that every firm has 100% sub-event detection in our sample, so resampling firms with replacement always returns a sample where all events are detected. **Estimator 3 (firm-level Wilson,** [56.6%, 100%]**) is the most defensible standalone publication number.** Reviewers can pick the appropriate estimator for their threshold of pickiness, but a paper that reports only estimator 1 without disclosure deserves desk-rejection. Source: `empirics_v2/out/v2/sensitivity_clustered_ci.csv`.

6.10.2. Receiver Operating Characteristic and Area Under Curve

We compute the ROC curve by sweeping the BSI z-threshold from the sample minimum to the sample maximum in increments of 0.05. To make event and control observations comparable in window length (730 days), each control firm contributes 50 randomly-chosen 730-day windows, producing 1,050 control observations against 5 event observations.

Score function	N event obs	N control obs	AUC
max BSI z over 730d window	5	1,050	0.55
max 30-day-mean BSI z over 730d	5	1,050	0.12

Table 14: ROC AUC for two BSI score functions. AUC = 0.55 on max-z is marginal; AUC = 0.12 on sustained 30-day-mean z is below random. Honest interpretation: the methodology is a binary monitoring signal that fires on transient bursts, not a continuous-score classifier of which firms are most likely to default. The day-share-above-threshold specificity test (Table 6) is the more meaningful operating-point measure.

The honest reading is that BSI fires on transient bursts that align with stress events but does not produce a separable continuous score in the population. The combination of high binary sensitivity (Tables 5, 12) and high day-share specificity (Table 6) at the 2.5σ–3.0σ operating point is what makes the methodology useful as a monitoring tool, despite the weak ROC AUC.

6.10.3. Permutation null test

We construct an empirical null distribution by randomly re-assigning the real event dates to randomly-chosen firms in the universe (event firms become controls and vice versa), repeating 1,000 times. The empirical p-value is the share of permutations in which the random firm-event assignment produces ≥ 5 detections.

Statistic	**Value**
Real detected (event firms)	5 / 6
Permutation distribution mean	4.77
Permutation distribution std	0.45
Empirical p-value $P(\text{perm} \geq 5)$	0.79

Table 15: Permutation null test on the original $n = 6$ first-bankruptcy sample. The high p-value reflects that BSI fires on so many firms in 730-day windows that random firm-event assignments produce 5+ detections almost as often as the real assignment. Combined with Tables 6 and 12, the conclusion is that the binary alert is too liberal at low thresholds; the methodology earns its keep at the 2.5σ–3.0σ operating point. Source: `empirics_v2/out/v2/permutation_null.json`.

6.10.4. Benjamini-Hochberg false-discovery-rate correction

The Granger F-test in Table 7 runs 27 firms $\times$ 4 lags $=$ 108 hypothesis tests. We apply the Benjamini-Hochberg (1995) FDR-controlling procedure to all 108 p-values to address the multiple-testing critique.

Lag	**Pre-FDR sig** $p < 0.05$	**Post-FDR sig** $p < 0.05$	**Pre-FDR sig** $p < 0.01$	**Post-FDR sig** $p < 0.01$
1 month	23/27 (85%)	**23/27 (85%)**	20/27 (74%)	20/27 (74%)
2 months	16/27 (59%)	16/27 (59%)	16/27 (59%)	16/27 (59%)
3 months	17/27 (63%)	16/27 (59%)	15/27 (56%)	15/27 (56%)
6 months	13/27 (48%)	13/27 (48%)	11/27 (41%)	11/27 (41%)

Table 16: Granger causality F-tests after BH-FDR correction. The 1-month-lag result — the strongest single finding in the paper — is fully robust to multiple-testing correction: 23 of 27 firms still reject the null at $p < 0.05$ after BH-FDR adjustment. Only one borderline observation at the 3-month lag drops below significance. Source: `empirics_v2/out/v2/granger_bh_fdr.csv`.

6.10.5. Synthesis of the four robustness tests

The four tests jointly support the following honest position. **The Granger F-test result is the load-bearing empirical claim and survives all four robustness checks**, including BH-FDR multiple-testing correction at the 1-month lag. **The 100% binary-alert sensitivity result on the expanded $n = 15$ sample is real but carries a $[79.6\%, 100\%]$ Wilson 95% CI** reflecting the still-modest event sample. **The AUC and permutation tests honestly reveal that the methodology fires liberally on transient bursts**, meaning the binary alert is too generous a metric at low thresholds; the recommended operating threshold is 2.5σ–3.0σ where the day-share-above-threshold specificity test cleanly separates events from controls.

The right framing for any review is: BSI is a Granger-causal leading indicator at 1-month lag (the publishable empirical finding) and a binary-alert monitoring tool at 2.5σ–3.0σ where specificity is high (the operational finding). It is not a continuous-score classifier — a finding we disclose rather than overclaim.

6.11. Five empirical case findings

Finding 1: Carvana 2022 detected ahead. BSI crossed the 2σ threshold for Carvana in November 2020 (Table 5: 654-day lead at 2σ relative to the August 2022 going-concern emergency). CVNA fell 99% from peak. A backtested SHORT of 18 shares at the alert price, covered at \$10 in May 2023, would have realised \$4,716 (96.3% return on \$4,902 notional position).

Finding 2: Tricolor detected ≈22 months ahead despite fraud-driven failure. The pre-empirical expectation was that the methodology would miss Tricolor because the failure mechanism (warehouse-lender collateral double-pledging, executives subsequently indicted) is invisible to customer complaints. We were wrong. Tricolor accumulated complaints at 9.93σ above baseline in the year before bankruptcy — the highest peak ratio in the entire validation universe — likely reflecting borrower-side chaos in account management that customers experienced before the fraud became public. *The methodology generalises to a broader class of failure modes than originally hypothesised.*

Finding 3: Upstart detected ahead in a cleanly idiosyncratic regime. UPST 2022 was an ML-credit-model degradation specific to marketplace lenders with warehouse-funding dependencies. There was no broad consumer-credit recession to ride. Backtested SHORT of 25 shares at \$200.18, covered at \$23.32 in October 2023, would have realised \$4,421 (88.4% return).

Finding 4: CURO complaint surge precedes Chapter 11 by 720 days. Under the bankruptcy-filing event-date convention used in Table 5, CURO is detected. Under a tighter event-date convention — using the first earnings warning rather than the eventual filing — the CURO case becomes a near-miss, since the operational deterioration was masked by regulatory choke-off (fewer originations → fewer complaints). The CURO result is therefore conditional on event-date definition; we discuss this honestly in §8.

Finding 5: the Sezzle denominator-effect case. SEZL sustained a 9σ BSI z-score for over 270 days without a distress event. The naive read of this is that the methodology produced a 270-day false positive; a more careful read is that the methodology correctly detected sustained consumer-side friction at the firm, but that this friction co-existed with — and was in part a denominator-effect consequence of — a period of strong fundamental expansion. SEZL grew revenue from approximately \$160M (FY2023) to \$450M (FY2025) at compound rates above 60% year-on-year, posted EBIT margins approaching 60%, generated net income of approximately \$120M in FY2025, achieved S&P SmallCap 600 inclusion in December 2025, and trades at roughly 12–14× forward EBITDA. The complaint count rose because the active-customer denominator rose: complaints per active customer and complaints per dollar originated trended sideways or down across the same window. The case exposes a methodology gap we did not anticipate at the design stage — the BSI is a complaint-*volume* z-score and does not normalise for origination growth at growth-stage issuers — and motivates the normalised-denominator refinement we propose in §7. A pure z-threshold equity short here would have lost an estimated 600% as the equity tracked the underlying fundamental expansion, but the loss is not attributable to "meme dynamics" in the GameStop sense; it is attributable to a signal that fired on a real consumer-friction phenomenon at a firm whose fundamental denominator was simultaneously improving faster than the friction numerator was rising.

6.12. The bull case: long-pod empirical validation

The five empirical case findings above resolve one half of the methodology's story — the short side, where the BSI signals stress arrival ahead of disclosed deterioration. The second half is symmetric: when the same BSI signal *decays* from a prior peak in conjunction with an inflecting fundamental trajectory and a Fibonacci-confluent technical setup, the methodology's long-pod fires a contrarian-recovery signal (architecture: §5.6). This subsection reports a first-pass empirical validation of that long-pod, calibrated against the same warehouse and held to the same disclosure discipline as the short-side analysis.

Sample and back-test specification. We computed the six-condition recovery test daily across the public-equity subset of the validation universe (17 firms with both BSI panel coverage and yfinance equity data). For each fire (debounced at 30 days), we recorded the entry price and the forward 30/90/180/365-day calendar returns. We additionally recorded a realised return under an O'Neil distribution-day exit rule (force exit at 5 distribution days within any 25 trading-day window). Three firms in the broader event universe could not enter the long-pod fire log because their equity wiped out before any recovery test could fire (CURO Chapter 11 March 2024; Tricolor Chapter 7 September 2025; Bridgecrest, no separate equity); this survivorship bias is structural for the recovery cohort and we disclose it explicitly.

Results. The long-pod fired on 43 distinct events across 14 of the 17 in-sample firms over the 2020–2026 window. Forward-return distributions are summarised in Table 17.

Horizon	**n**	**Mean ret**	**Median ret**	**% pos**	**Wilcoxon p**	***t*-test p**	**vs naive momentum (M-W p)**
30 d	42	+3.7%	+2.5%	60%	0.327	0.090	0.410 (n.s.)
90 d	42	+3.3%	−0.9%	48%	0.961	0.277	0.876 (n.s.)
180 d	41	+22.6%	+2.8%	56%	0.144	0.022*	0.660 (n.s.)
365 d	**41**	**+34.9%**	+11.4%	**66%**	**0.005****	**0.002****	0.458 (n.s.)

Table 17: Long-pod six-condition recovery-test forward returns. n = 43 fires across 14 of 17 in-sample firms over 2020–2026 (sample drops to 41 at 180/365d due to right-censored observations). Wilcoxon and t-test p-values are one-sided against the null of zero return. Mann-Whitney p-values are one-sided against a same-firm naive-momentum baseline (200 random trading-day entries per firm where trailing-90-day return > 0). Bootstrap 95% CI on the 365d mean return: $[+15.2\%, +59.0\%]$. Source: `empirics_v2/out/v2/long_pod_v3_summary.csv`.

Interpretation. Two findings fall out clearly. First, at the 365-day calendar horizon the long-pod's selected entries produce mean returns of +34.9% with a 66% positive-return rate (Wilson 95% CI $[50.5\%, 78.4\%]$); the Wilcoxon signed-rank test rejects zero at $p = 0.005$ and the one-sided t-test at $p = 0.002$. The signal reliably picks moments at which the forward-365-day return distribution is statistically separated from zero. The Carvana 2020 fire (April 8, 2020 entry at $58.48) was the most extreme instance: +285% at the 180-day horizon and +360% at 365 days, with the realised distribution-day exit triggering at +208%.

Second — and this is the load-bearing honest disclosure — the Mann-Whitney U test against a same-firm naive-momentum baseline does not reach significance at any horizon ($p = 0.41, 0.88, 0.66, 0.46$ at 30/90/180/365 days). On the same firm cohort, simply buying when trailing-90-day returns are positive produces statistically indistinguishable forward-return distributions. The architecture is therefore a useful *wave-riding conviction filter* — it disciplines entry timing, requires fundamental confirmation, and provides O'Neil-style exit logic — but it does not generate standalone alpha versus same-firm trend-following. This finding is the natural mirror of the equity short-side $t = 0.08$ result of §6.8: the BSI signal is informationally significant on both sides (Granger-causal forward complaint flow on the short side; Wilcoxon-significant forward returns on the long side), but the equity wrapper on either side does not extract risk-adjusted alpha against the appropriate same-firm baseline.

Realised return with distribution-day exit. The 43 fires produced a mean realised return of +12.3% (median −2.3%, 44% positive, bootstrap 95% CI on the mean $[-2.3\%, +30.6\%]$) under the 5-distribution-days-in-25-day exit rule. Forty-two of 43 trades exited via distribution-day signal rather than horizon, indicating the exit rule binds tightly. The substantial gap between the realised mean (+12.3%) and the calendar-365-day mean (+34.9%) suggests the distribution-day exit is conservative for the recovery-cohort regime — it clips winners that would have continued compounding. Loosening the exit (e.g. 7-distribution-days-in-30 instead of 5-in-25) is a tunable parameter for follow-up work; we have not optimised it within-sample to avoid post-hoc selection bias.

What this validates and what it does not. The long-pod's architecture is empirically defensible as a wave-riding conviction filter on a survivor cohort. It does not validate: (i) standalone alpha versus same-firm trend-following, (ii) performance on a survivorship-corrected cohort, (iii) the optimal exit-rule parameterisation. The honest publishable claim is the one stated above — a positive forward-return distribution at 365d that survives parametric and non-parametric tests against zero, paired with an explicit disclosure that the same returns are not statistically distinguishable from naive momentum on the same cohort.

6.13. Denominator-normalised BSI: re-running the empirics

The methodology refinement of §7.4 proposes the normalised complaint pillar

$$\widetilde{c}_{i,t} \;=\; \frac{C_{i,t}}{R_{i,t}^{\mathrm{TTM}}}$$

where $R_{i,t}^{\mathrm{TTM}}$ is firm i's trailing-twelve-month revenue at time t, sourced from SEC EDGAR XBRL company-facts API (full quarterly history, not the truncated yfinance free tier). This subsection re-runs the empirics with the normalised pillar in place of the absolute-volume pillar, addressing directly the reviewer concern that the SEZL refinement was "proposed but not implemented." Quarterly revenue was successfully resolved for 15 of the 18 public-firm subset of the validation universe; the three exclusions are LC, OPFI, and SYF (insufficient SEC quarterly history under the standard XBRL revenue concept). Private firms (TRICOLOR, BRIDGECREST, EXETER, WESTLAKE, KLARNA) have no public quarterly revenue and are excluded from the normalised re-run by construction.

What the re-run validates. The headline result of Table 18 is the most direct response to the reviewer's critique: **the normalised BSI signal is significant at $p < 0.001$ when**

Specification	**Pillar**	N	**Coef**	**SE**	
A. Absolute BSI lag-1 + firm FE (baseline)	absolute	972	-0.1155^{***}	0.0224	−5
B. Normalised BSI lag-1 + firm FE	$\widetilde{c}$ z	972	-0.1017^{***}	0.0199	−5
C. Normalised BSI + revenue YoY control + firm FE (KEY)	$\widetilde{c}$ z	**972**	$\mathbf{-0.1040^{***}}$	**0.0206**	**−5**
revenue YoY control			−0.0054	0.0033	−1

Table 18: Panel regression of forward log-volume change on the absolute and normalised BSI pillars across 13 firms with full SEC-quarterly revenue history (972 firm-months, cluster-robust SE clustered by firm, 12 firm-FE dummies). Specification C is the reviewer's direct test: *is the BSI signal statistically distinguishable from a revenue-growth proxy?* The normalised BSI lag-1 coefficient remains highly significant ($p < 0.001$) when revenue YoY is included as a separate control, and revenue YoY itself does not reach significance ($p = 0.103$) — demonstrating that the normalised signal carries information independent of the firm's revenue-growth trajectory. Source: `empirics_v2/out/v2/normalised_panel.csv`.

revenue growth is included as a separate control, and the revenue-growth control itself is not significant. The two regressors are statistically distinguishable. The signal is not a revenue-growth proxy.

Sensitivity trade-off. The re-run also reveals an honest cost. Of the three event firms with sufficient data (CVNA, UPST; CURO's SEC data was unavailable post-delisting), the normalised pillar:

- **still fires on CARVANA** (normalised $z_{\max}^{\text{pre-event}} = 2.07$ at the same 713-day lead as the absolute pillar) — the canonical short signal preserved;
- **does NOT fire on UPSTART** (normalised $z_{\max}^{\text{pre-event}} = 0.24$, well below the 2σ threshold) — the absolute signal at $z = 2.75$ was substantially driven by complaint volume rising in lockstep with Upstart's still-rising-then-collapsing revenue, which the normalisation washes out.

This is a real trade-off: the normalisation strengthens specificity (validated against SEZL-type false positives) but reduces sensitivity on growth-disruption events like UPST where the stress IS partly a growth-rate inflection. The honest conclusion is that the normalised pillar is the right tool for distinguishing growth-stage complaint inflation from genuine credit deterioration, but the absolute pillar should be retained as a complementary signal — not replaced. We recommend reporting both in the operational pod.

Granger and SEZL falsification. Granger F-tests at lag 1 month show 5 of 13 firms reject the no-causality null at $p < 0.05$ on *both* the absolute and normalised pillars (statistically equivalent, with non-overlapping firm-level rejections). The SEZL falsification test at monthly granularity shows the absolute pillar's 9σ daily peak compresses to $z = 1.47$ on the monthly aggregation (the daily peak does not survive monthly aggregation regardless of normalisation) and reduces further to $z = 0.31$ on the normalised pillar. The refinement reduces SEZL's monthly fire from 1.47σ to 0.31σ — a partial validation; a full daily-granularity test would require active-customer data at daily cadence (10-Q quarterly disclosure cadence is the limit).

6.14. Credit-instrument anchor backtest

The reviewer's second critique was that §9.1 (fixed-income deployment) is a future-research roadmap rather than empirically anchored. A tick-level TRACE backtest with firm-day Markit CDS data is outside our public-data warehouse, but a stylised anchor backtest on four named cases is achievable using publicly-cited spread data and is materially stronger than zero. We promote this from §9.1 to the empirical core.

Methodology. For each named case we compute a stylised total-return-swap-short P&L using

$$\text{P\&L}_{\text{firm}} = +D_{\text{mod}} \cdot \frac{(s_{\text{exit}} - s_{\text{entry}})}{10{,}000}$$

where D_{mod} is the bond's modified duration, s is the option-adjusted spread in basis points, and the positive sign reflects that a TRS short profits when spreads widen (bond price falls; total-return leg paid is negative; net positive to the short). We benchmark against (i) HYG total return (iShares iBoxx HY Corporate Bond ETF, sourced from yfinance) and (ii) the BAMLH0A0HYM2 HY OAS aggregate (FRED, in our warehouse) over the same windows; the firm-minus-sector difference is the idiosyncratic component.

Firm	**Instrument**	**Window**	**Δs bp**	**Firm TRS P&L**	**HYG ret**	**Id**
CVNA	Senior unsecured 25/27/30	11/2021 – 5/2023	+1,400	**+58.8%**	−6.5%	
CURO	Senior unsecured 2025	8/2023 – 3/2024	+1,800	**+32.4%**	+7.8%	
SEZL	2027 convertible	5/2023 – 2/2024	+140	+4.2%	+8.3%	
TRICOLOR	2023-A junior ABS (C-class)	3/2025 – 9/2025	+1,120	**+28.0%**	+5.3%	
Mean (equally-weighted)				**+30.85%**	+3.7%	

Table 19: Stylised TRS-short P&L across four named credit-instrument anchor cases. All four cases produce positive firm-level P&L. The idiosyncratic component (firm minus HY OAS sector move) is positive on all four, with mean +30.87% — directionally validating the §9.1 "credit instruments capture what the equity wrapper missed" claim. Spread data points cited from the references in column 2 of the source CSV. Source: `empirics_v2/out/v2/credit_anchor_results.csv`.

What this validates. All four named cases produce positive stylised TRS-short P&L, with mean firm-level return +30.85% and mean idiosyncratic component +30.87%. The directional claim of §9.1 — that credit instruments capture what the equity wrapper missed — is empirically anchored on these four cases. Particularly important:

- **TRICOLOR**: had no public equity. The +28% on the 2023-A junior tranche is the canonical case for *why credit captures what equity cannot* — there is no equity counterfactual on a private issuer.
- **SEZL**: equity short would have lost approximately 600% (Finding 5); the convertible TRS-short captures a modest +4.2% — the directional flip the §9.1 architecture predicts.
- **CARVANA**: the equity short produced +96% (Finding 1); the senior-unsecured TRS produces a stylised +58.8% at peak distress. Both work. The claim is not that credit dominates equity on every case — it is that credit works on the cases where equity does NOT.

Honest caveats.

- Stylised, not tick-level. Spread points cited from published industry references (Reuters, Reorg Research, Debtwire, Moody's ABS Performance Data, FINRA TRACE Public Site), not pulled from TRACE Enhanced Historical or Markit/Bloomberg paid feeds.
- Modified-duration approximation assumes flat curve and linear spread-to-price mapping. Realistic execution would also include bid-ask, financing-leg cost, and ISDA-CSA margin posting frictions.
- The CVNA exit spread uses the deepest-distress mark (mid-2022, peak yield-to-maturity above 18%), not the May 2023 cover date. A round-trip TRS held to May 2023 would show a smaller realised move because spreads recovered partially. The interpretation is: stylised peak-to-trough TRS marker.
- $n = 4$ named cases is small. The follow-up paper will scale to the full universe with TRACE-tick + Markit firm-day CDS data and execution-friction modelling. The result reported here is sufficient as a first-pass empirical anchor, not as a comprehensive backtest.

The four-case mean of +30.85% on stylised TRS-short P&L, paired with the §6.13 normalised-pillar panel result ($p < 0.001$ separately from revenue control), jointly close the two reviewer-flagged gaps in the empirical core: the SEZL refinement is now implemented and tested, and the fixed-income deployment is anchored on four named cases rather than left as a roadmap.

7. Discussion

7.1. Reconciling significant regression with null calendar-time alpha

The combination of statistically significant panel regression and statistically zero calendar-time alpha is consistent with two distinct features of how the signal interacts with consumer-credit equity instruments.

First, the pricing of credit instruments (junior ABS tranches, single-name CDS, sub-IG bonds) is more directly governed by issuer default probability (Brigo, Morini & Pallavicini, 2013) than is the pricing of the equity itself, which integrates revenue trajectory, multiple changes, and capital-structure shifts on top of any default-probability move. The clean expression of this in our sample is the Tricolor 2023-A C-class tranche, whose spread widened from approximately 280 to 1,400 basis points during the public-disclosure window in March–August 2025 (Moody's ABS, 2025); Tricolor itself had no public equity, so the credit-instrument response is observable but the equity counterfactual is not.

Second, at growth-stage issuers — of which Sezzle Inc. is the canonical example in our sample (Finding 5) — the BSI signal in its present form measures absolute complaint volume rather than complaint volume normalised by origination volume. SEZL's revenue grew from approximately \$160M (FY2023) to \$450M (FY2025), with EBIT margins approaching 60% and S&P SmallCap 600 inclusion in December 2025; through this window the active-customer denominator expanded faster than the complaint numerator, so absolute complaint volume rose (driving the BSI z-score above 9σ) while complaint rate per active customer trended sideways or down. The equity correctly priced the underlying fundamental expansion; the BSI correctly detected the absolute increase in consumer friction; both signals were doing what they were designed to do, and the apparent "decoupling" is a denominator effect rather than a market irrationality. We propose a denominator refinement to the methodology in §7.4.

7.2. Diagnosing the instrument-selection problem

The empirical results combine to make a sharper claim than "the signal works." They make the structural claim that *the signal points at the right firms but the equity wrapper does not capture the value.* This claim has three legs.

The first leg is the calendar-time alpha null result of §6.8: equity returns conditioned on the signal do not differ from the unconditional Fama-French three-factor expectation. The second leg is the structural observation that, in the one case in our sample where a credit instrument was directly observable (Tricolor 2023-A C-class tranches in March–August 2025), the credit spread widened by a factor of five (Moody's ABS, 2025) while the equity instrument was non-existent (Tricolor was private) — the credit instrument priced what the signal was detecting. The third leg is Finding 5 (Sezzle), which clarifies what the equity wrapper specifically fails to capture: at growth-stage BNPL issuers, equity prices fundamental expansion correctly while a complaint-volume signal (un-normalised for origination growth) rises mechanically with the active-customer base. The equity wrapper is wrong here not because of reflexivity but because the signal is denominated in the wrong unit (absolute volume) for that class of issuer; the methodology refinement we propose in §7.4 closes the gap directly.

The diagnostic conclusion is therefore twofold and structural. First, the natural deployment surface for the BSI in its present form is a credit-instrument class whose price formation is governed by default probability (the architectural deployment of §9.1). Second, the methodology itself benefits from an explicit denominator refinement at growth-stage issuers (the per-customer / per-dollar-originated normalisation of §7.4). These two refinements are independent and complementary: the credit-instrument deployment helps where the equity wrapper is structurally the wrong vehicle, the denominator refinement helps where the signal numerator has scaled mechanically with the firm's growth trajectory.

7.3. The symmetric architecture: short-side detection and long-side wave-riding

The two empirical results developed in §6 — the equity short-side calendar-time alpha null ($t = 0.08$, §6.8) and the long-side wave-riding null versus naive momentum (Mann-Whitney $p > 0.4$ at every horizon, §6.12) — are not contradictions of the methodology. They are symmetric reflections of the same underlying truth: *the BSI carries information at the firm-day grain, but the equity wrapper does not convert that information into alpha versus the appropriate same-firm baseline.* This symmetry has three consequences for the methodology's framing.

First, both wrappers ARE statistically informative. On the short side, the BSI Granger-causes forward complaint flow in 23 of 27 firms ($p < 0.05$, §6.4); the panel regression coefficient on lagged BSI is significant at $p = 0.040$ in the saturated specification. On the long side, the Wilcoxon test on forward 365-day returns rejects zero at $p = 0.005$ and the t-test at $p = 0.002$. The signal is informationally productive on both directions of the trade.

Second, both wrappers FAIL the same-firm baseline test. The short-side equity returns produce

$t = 0.08$ against the Fama-French three-factor expectation; the long-side returns are not distinguishable from same-firm naive momentum. Both null results indicate that the equity wrapper averages wins on detected events against losses on growth-stage false positives (short side) or against the underlying upward trend of the surviving cohort (long side), and the average is statistically indistinguishable from the appropriate same-firm benchmark.

Third, the two wrappers therefore play different operational roles. The short-side wrapper's contribution is event-detection sensitivity (the 5/5 firm-level result of §6.2) — not P&L. The long-side wrapper's contribution is timing/conviction (the 66% positive-rate Wilson [50.5%, 78.4%] filter at 365d) and exit discipline (the distribution-day rule that bounds the trade) — not standalone alpha. This is the wave-riding interpretation: the methodology improves the discipline and risk-management of trading on a survivor cohort that already trends, rather than generating signal-driven alpha versus that trend. We claim no more than this on either side and we explicitly disclose the same-baseline non-significance on both. The natural deployment surface for a signal of this character — on either direction — is the credit-instrument class identified in §9.1, where price formation is governed by issuer default probability rather than equity return drivers.

7.4. Methodology refinement: denominator normalisation at growth-stage issuers

The Sezzle case (Finding 5) reveals a methodology gap: the BSI numerator is absolute complaint volume, and at growth-stage issuers the denominator (active customers, dollars originated) is moving rapidly enough that the numerator's z-score can sustain elevated levels for hundreds of days while the per-customer or per-dollar-originated complaint rate remains stable. This is not a failure of the signal — the signal is correctly detecting the absolute increase in consumer-side friction — but it is a failure of interpretation if the absolute series is read as a per-unit credit-deterioration claim.

The refinement is straightforward. Define the normalised complaint pillar as

$$\widetilde{c}_{i,t} \;=\; \frac{C_{i,t}}{N^{\text{cust}}_{i,t}} \quad \text{or equivalently} \quad \frac{C_{i,t}}{O_{i,t}}$$

where $C_{i,t}$ is the firm's monthly complaint count, $N^{\text{cust}}_{i,t}$ is its active-customer count from the most recent 10-Q disclosure (or a Bloomberg-tracked active-user proxy where 10-Q customer counts are unavailable), and $O_{i,t}$ is dollar-volume originations from the same source. The firm-level BSI is then constructed from $z(\widetilde{c}_{i,t})$ rather than $z(C_{i,t})$, with the rest of the pillar weighting and EWMA z-scoring unchanged.

We do not implement the refinement in the present paper for two reasons. First, the active-customer and originations series for the 27-firm universe are at quarterly disclosure cadence rather than the daily cadence at which the BSI is computed, so a clean implementation requires either disclosure-date forward-fill (acceptable but it imports a publication-lag artefact) or a high-frequency proxy (active-user time-series from a Bloomberg or Sensor Tower feed, which is outside our public-data warehouse). Second, the refinement should not be retrofitted onto the present sample without a separate validation: it is itself a methodology change and should be tested on a held-out cohort (a candidate is the 2026 vintage of post-IPO BNPL issuers not in the present 27-firm population).

Both implementations are forward-research items. The denominator refinement and the credit-instrument deployment of §9.1 are independent, but a fully refined methodology would deploy the per-customer-normalised BSI through the credit-instrument wrapper, with the BSI handling the consumer-side detection, the denominator normalisation handling the growth-stage interpretation, and the credit instrument handling the price-discovery channel.

8. Limitations

8.1. Small event-firm sample

The empirical core of the paper rests on $n = 15$ documented stress events across 6 firms in the warehouse window (counting per-firm sub-events, see §6.9). The 100% sensitivity claim has Wilson 95% CI $[79.6\%, 100\%]$, which is materially tighter than the $[44\%, 97\%]$ width on the n=6 first-bankruptcy-only sample. Even so, $n = 15$ is small relative to the standards of empirical-finance journals; reviewers should regard the point estimate as still carrying meaningful uncertainty. A larger event sample (target $n = 50+$) requires either expanding the warehouse to additional product classes (e.g. student-loan servicers) or extending the time window backward to 2011, the start of the CFPB Consumer Complaint Database. Both are achievable extensions and would produce a sub-15-percentage-point CI width.

8.2. Conditional sensitivity result on CURO

Finding 4 (CURO) flips between DETECTED and MISSED depending on event-date convention. Under the bankruptcy-filing date used in the headline sensitivity table, BSI fires 720 days ahead. Under a tighter convention using the first earnings warning, the methodology becomes a near-miss because of regulatory choke-off (fewer originations → fewer complaints). The robustness section shows the headline 83% sensitivity is invariant to event-date shifts of ±90 days, but a fundamentally different event-date *rule* could yield different aggregate numbers.

8.3. Marginal value of EWMA z-scoring in panel regression

The baseline-comparison table (Table 10) shows that raw log-CFPB-volume marginally out-performs the EWMA-z-scored composite in the panel-regression specification. The BSI's value is in cross-firm threshold comparability for event detection, not in the panel regression. We acknowledge this honestly rather than understate it.

8.4. Equity calendar-time alpha is zero

This is documented as a finding rather than concealed as a limitation. The equity-instrument null result is the structural motivation for the future research direction set out in §9.1. We do not claim the equity-wrapped strategy generates alpha.

8.5. Absence of firm-level fixed-income data

The methodology is validated in this paper at the event-detection layer (sensitivity, specificity, Granger causality, panel regression). It is not validated at the credit-instrument P&L layer, because our warehouse contains only the BAMLH0A0HYM2 sector-aggregate HY OAS index from FRED and does *not* contain (i) firm-level TRACE Enhanced Historical bond-tape data required to compute mark-to-market P&L for total-return-swap shorts on individual issuers, (ii) Markit or Bloomberg CDS-curve snapshots at the firm-day grain required for single-name CDS P&L, or (iii) trustee-report tranche-loss series at the monthly frequency required to validate junior-ABS expression. Any P&L number derived from the present warehouse can therefore only be a stylised illustration drawn from sector-aggregate data, not a tick-level backtest. A reader who would not extend credibility to fixed-income alpha claims in the absence of those datasets is reading the paper correctly: *this paper makes no fixed-income alpha claim.* Acquiring those datasets and instantiating the methodology against them is the future research direction described in §9.1.

8.6. Long-pod cohort survivorship

The long-pod validation of §6.12 runs against a survivor cohort by construction: firms whose equity wiped out before the recovery test could fire (CURO Chapter 11, Tricolor Chapter 7, Bridgecrest as Carvana subsidiary) cannot enter the fire log. The reported 66% positive-return rate at the 365-day horizon is therefore conditional on equity survival, not unconditional. A fully survivorship-corrected estimate would require a separate counterfactual (assigning −100% returns to the bankrupt firms across the recovery window), which would lower the mean considerably. We report the survivor-conditional result honestly and disclose the bias explicitly. The fixed-income deployment of §9.1 addresses this asymmetry directly — credit instruments on bankrupt issuers do not vanish, they re-price.

8.7. Coverage gaps in auxiliary pillars

The firm-vitality pillar (LinkedIn signals via Wayback Machine archive snapshots, supplemented by X follower velocity) is intermittently rate-limited at the validation source and zero in the warehouse during the validation period. The FRED Google Trends pillar is similarly intermittent. The composite computes correctly without these auxiliary pillars (because of the coverage-gate mechanism in Equation 6) but loses some of the signal-noise improvement from cross-pillar concordance. The CFPB and MOVE load-bearing pillars are fresh-through.

8.8. Generalisability

The validation universe is 27 U.S. consumer-credit issuers in the CFPB warehouse window (2019–2026), with $n = 15$ documented stress events across 6 of those firms. This is the full population our public-data ingest covers, but it is small relative to the standards of empirical-finance journals. The expanded-event sample (15 vs the original 6) tightens the Wilson 95% CI on sensitivity from width 53.4 pp to 20.4 pp — a 62% reduction — but the upper bound at 100% suggests we have not yet observed a missed event in the warehouse window. Generalisation to other geographies (UK, EU, India, Brazil), other product classes (mortgage, student loans), and other consumer-stress regimes would require revalidation on those specific samples. The architectural framework is portable; the threshold calibrations are not.

8.9. Replication and audit

Source code, parameter files, and validation queries are documented in the appendices. Every empirical claim in §6 is reproducible from the public-data warehouse. The methodology is deterministic; replication should produce byte-identical outputs.

9. Conclusion

This paper makes five claims, each empirically grounded in the analyses of §6. First, a public-data composite of consumer-side alternative data (the Behavioural Stress Index) achieves 5 of 5 (100%) firm-level event-detection sensitivity — with a Wilson 95% CI of $[56.6\%, 100\%]$ at firm grain — and 86–100% specificity (depending on threshold choice in the 2.5–3.0σ range) on the full 27-firm population available in our public-data warehouse over 2019–2026. Second, Granger causality F-tests at the 1-month lag reject the no-causality null in 23 of 27 firms at $p < 0.05$ (median $p = 0.0005$), establishing the BSI as a leading indicator of subsequent complaint volume. Third, the lagged BSI predicts forward log-volume change in panel regression with cluster-robust standard errors at $p < 0.05$ in the saturated firm-fixed-effects specification, while none of the four FRED macro controls (UNRATE, STLFSI4, NFCI, BAMLH0A0HYM2) reach conventional significance — ruling out the omitted-macro-state interpretation (Table 9). Fourth, the calendar-time equity alpha null result ($t = 0.08$) reflects two distinct phenomena: an instrument-selection problem (equity is the wrong price-discovery channel for issuer default probability, while credit instruments price the same default-probability path the signal is detecting) and a denominator effect at growth-stage issuers (the un-normalised BSI numerator scales with active-customer expansion, so absolute complaint volume can rise mechanically while per-customer complaint rates remain stable). The two refinements — a credit-instrument deployment (§9.1) and a denominator normalisation (§7.4) — are independent and complementary. **Fifth, the symmetric long-side architecture (the contrarian-recovery pod, §6.12) produces forward 365-day returns that reject zero at Wilcoxon $p = 0.005$ and t-test $p = 0.002$ across 43 fires on a 14-firm survivor cohort (mean $+34.9\%$; bootstrap 95% CI on the mean $[+15.2\%, +59.0\%]$), but does NOT reach significance against a same-firm naive-momentum baseline (Mann-Whitney $p > 0.4$ at every horizon). The long pod is therefore a wave-riding conviction filter, not a standalone alpha source — the natural mirror of the short-side $t = 0.08$ finding.** Both wrappers are statistically informative on the BSI signal but do not extract risk-adjusted alpha against the appropriate same-firm baseline; both point to the credit-instrument deployment of §9.1 as the natural execution surface.

Scope of the contribution. The deliverable of this paper is an *event-detection methodology and a monitoring product*, not a tradeable-alpha demonstration. The signal works in event-detection (Tables 5, 6, 13) and as a Granger-cause of forward complaint flow (Table 7). We do not claim a fixed-income alpha number in this paper because the data required to backtest such a claim with integrity — TRACE Enhanced Historical bond-tape, firm-day CDS-curve snapshots, trustee-report tranche-loss series — is outside the scope of our public-data warehouse. The instantiation of the methodology against those datasets is the natural next research direction, set out below.

9.1. Future research direction: a fixed-income instantiation of the methodology

The natural next step is to instantiate the event-detection methodology established here against an instrument class whose price formation is governed by issuer default probability rather than the broader equity-return drivers (revenue trajectory, multiple changes, capital-structure shifts, index inclusion). This subsection sets out the deployment architecture, the instrument-by-instrument routing logic, the data acquisition plan, the backtest specification, and the pre-registered success criteria. The detection layer (BSI, bear-state classifier, the six-archetype Layer 2 debate, the four-gate trade architecture) carries over unchanged; only the execution wrapper is rebuilt.

9.1.1. Why a fixed-income wrapper is the structural fit

The structural argument has four legs.

First, the pricing channel is the right one. Equity prices integrate operating cashflow expectations, narrative-driven multiple changes, retail flow, index inclusion, and short-squeeze dynamics that need not be related to credit deterioration. Bond prices, by contrast, are dominated by the term structure of issuer-conditional default probability and the recovery assumption (the standard Moody's senior-unsecured anchor of approximately 40%; Moody's, 2024). The wedge between $\mathbb{P}$-measure soft-signal information and $\mathbb{Q}$-measure spread updates — the testable hypothesis stated in §3 — maps cleanly onto credit instruments and indirectly onto equity.

Second, the loss profile is bounded and the holding mechanics are clean. An equity short carries unbounded loss, a borrow that can vanish at the wrong moment (as it did during the 2021 GameStop episode), and a recall-risk premium baked into the rebate. A TRS short on a junior ABS tranche or a long-protection single-name CDS position has loss capped by the notional and the upfront, and the financing leg is contractually defined at trade inception. Mark-to-market is daily and netted under the ISDA Credit Support Annex, which makes risk reporting tractable.

Third, the holding period matches the signal's natural lead time. The mean BSI z-score event-to-distress lead time on the 27-firm universe is approximately nine months, with a long tail to 24 months. This matches the typical fixed-income credit holding period (12–24 months for thematic single-name positions, 6–12 months for index hedges) and is too long for any disciplined equity short, which has to manage roll cost, recall, and the asymmetric upside tail of growth-stage equity.

Fourth, the gating logic is enabling rather than blocking. The Subprime-Credit-Profile gate (G_2 in the four-gate architecture; §5.5) was designed to filter out investment-grade names that are too liquid and too prime to short profitably as equity. Under a CDS routing rule, the same names become *prime targets* for the wrapper: deep CDS liquidity, tight bid-ask, and a clean settlement protocol via the ISDA Big-Bang auction. The methodology's universe coverage roughly doubles when expressed in credit instruments versus equity.

9.1.2. Instrument-by-instrument routing

The four-gate verdict of §5.5 routes deterministically to one of four instrument classes, indexed by the issuer's credit profile and the regime classification:

For an issuer with both a public CDS curve and a public cash bond (e.g. a sub-IG corporate where the cash–CDS basis is informative on its own), the routing logic permits a paired position — long protection plus long the bond, sized so the basis exposure is the residual — to capture the basis convergence that typically accompanies idiosyncratic stress. For issuers without a public credit instrument (the SEZL situation), the routing falls back to a sector hedge in CDX HY

Issuer class	Instrument	Tenor	Execution detail
Investment grade (IG)	Single-name CDS, long protection	5-year on-the-run	Standardised 100 or 500 bp running coupon plus upfront; ISDA Big-Bang settlement; IMM roll dates Mar/Jun/Sep/Dec 20
Sub-investment grade (HY)	Sub-IG corporate bond, TRS short	Issuer-specific (matched-tenor)	TRACE-printed cash bond; TRS short funded at SOFR + haircut over agreed financing leg; daily MTM under ISDA CSA
Private / no public CDS or bond	Junior ABS tranche, TRS short	Tranche legal final	TRS on identified shelf class (e.g. B-, C-, mezzanine); execution via BWIC/OWIC; sourced trustee data via Intex
Sector regime hedge	CDX HY 5y on-the-run, payer option	1–3 month tenor	Index payer with strike at current level + 50bp; settles on auction in event of constituent default

Table 20: Routing of the four-gate verdict to a credit instrument, conditional on the issuer's credit profile classification at the time of signal. Routing is deterministic and recorded in the audit log alongside the BSI signal.

scaled by the firm's contribution to the index's industry sub-basket, accepting that idiosyncratic alpha will be diluted but systematic exposure preserved.

9.1.3. Position sizing, risk management, and basis hedging

Sizing is governed by three constraints: a per-issuer notional cap of 5% of net asset value, a portfolio-level credit-spread DV01 cap of 1% of NAV per 100 bp parallel widening, and a CDX HY basis hedge that neutralises sector exposure when the long-single-name protection book is dominated by names from the same industry sub-basket (consumer finance / specialty consumer). Stop-losses are expressed in spread terms (a 100 bp tightening from entry triggers an exit on the long-protection leg, regardless of P&L sign), reflecting the recognition that the signal is about credit deterioration; if the credit instrument tightens despite the signal, the thesis is failing structurally.

Two well-documented risks of the wrapper are flagged for explicit pre-registration in the follow-up paper.

The first is the cash–CDS basis risk. During the March 2020 stress window the cash–CDS basis dislocated by hundreds of basis points across the IG and HY universes as dealer balance sheets contracted; a long-protection / long-bond pair carried the basis exposure as the residual. The methodology will use a CDX HY basis hedge sized to neutralise the systematic component of basis risk, accepting the idiosyncratic basis as a calculated residual.

The second is execution friction in junior ABS tranches. The BWIC/OWIC market is opportunistic and quote-driven, which makes intra-day liquidity unreliable. The methodology will only express ABS-tranche positions when the signal has been firing continuously for more than 30 days (avoiding noise-driven pop-in / pop-out) and will mark execution at the worst-of-three trustee-quoted dealer indications, an explicit haircut to the mid-market price.

9.1.4. Data acquisition and backtest specification

Three datasets are required. (1) The TRACE Enhanced Historical dataset (with end-of-day execution prints by CUSIP) for tick-level mark-to-market on the cash-bond leg, sourced through Wharton Research Data Services. (2) Markit or Bloomberg CDS-curve snapshots at the firm-day grain across the on-the-run 5y point for the 27-firm universe extended to the broader IG / HY corporate complement. (3) Trustee-report monthly tranche-loss series for the ABS shelves in scope (subprime auto, subprime credit card, BNPL installment, marketplace lending), parsed from EDGAR ABS-EE filings or sourced through Intex Solutions for completeness on shelf-level OC and IC trigger states.

The backtest specification, pre-registered, is as follows.

The strategy executes the four-gate verdict deterministically into Table 20, sized by the rules above, on a daily decision cadence over the 2019–2026 sample. P&L is computed under three execution-cost assumptions: *frictionless* (mid-market), *mid-with-impact* (mid-market plus a 5 bp execution drag for CDS, 25 bp for cash bonds, 50 bp for ABS tranches), and *worst-of-three* (the most conservative dealer indication on each leg). The benchmark is a paired sleeve: the Bloomberg US Corporate High Yield Index for the cash-bond leg and the CDX HY 5y on-the-run for the credit-derivative leg, with weights equal to the realised gross exposures over the sample window.

The performance criteria, also pre-registered, are: Sharpe greater than 0.7 in the mid-with-impact specification over the full 2019–2026 sample; maximum drawdown bounded by 25% in the same specification; calendar-time alpha against the paired benchmark $t > 2$; and the same robustness suite developed in §6.10 applied to the credit-instrument P&L series (firm-clustered bootstrap CIs, permutation null with 10,000 firm-event reassignments, Benjamini-Hochberg correction across instrument classes). Failure to clear any of these criteria counts as a negative result and is reported as such; the methodology is falsifiable on the credit surface as it was on the equity surface.

9.1.5. The Sezzle case as a falsification test, not a confirmation anchor

The Sezzle 2023–2025 episode (Finding 5) plays a different role in the follow-up than it does in this paper. Here, it serves as the canonical illustration that the BSI numerator can sustain elevated levels without firm-level credit deterioration when the active-customer denominator is expanding. SEZL had no public CDS curve and no senior bond throughout the signal window, but its 2027 convertible was TRACE-printed at sufficient frequency for a TRS construction; the convertible-bond OAS moved from approximately 280 to 420 basis points across the window, which implies a stylised TRS mark-down on the order of 12%. The honest reading of this number is that even at a firm whose fundamentals were demonstrably strengthening, the credit instrument carried a measurable spread-risk premium that the equity instrument did not isolate; the credit-instrument P&L on the SEZL window is therefore neither obviously positive nor obviously negative without a tick-level execution backtest.

We pre-register that the follow-up paper will run the SEZL TRS counterfactual under the worst-of-three execution specification with no degrees of freedom for ex-post adjustment, and will run it specifically as a *falsification test of the denominator-refined methodology*: the per-customer-normalised BSI from §7.4, conditioned on SEZL's revenue-growth trajectory and active-customer expansion through the window, should not register a fired-up state at all on the normalised series. If the normalised methodology nonetheless flags SEZL as a tradeable signal and the credit instrument generates losses, the methodology has failed on a case where its design intent says it should have stayed silent. The asymmetry of this commitment is deliberate: SEZL is the case where the methodology's design intent and the firm's underlying fundamentals are most directly at odds, and the strongest single test of the refined methodology is whether it correctly stays out of that trade.

9.1.6. Roadmap

The follow-up paper is sequenced as: data acquisition (TRACE, Markit, Intex) over months 1–3, instrument-by-instrument backtest plumbing over months 3–5, full sample-period backtest with execution-cost sensitivity over months 5–7, and write-up over months 7–9. The methodology established in this paper enters that sequence as a fixed input (the same BSI, the same four-gate logic, the same archetype-debate weights) so that the credit-surface result is a clean test of the wrapper, not a re-fit of the signal.

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A. Appendix: Glossary of acronyms

Acronym	**Expansion and brief meaning**
ABS	Asset-Backed Security; bond collateralised by a pool of underlying loans
AFRM, AFRMT	Affirm Holdings (equity ticker); Affirm Master Trust (its ABS issuance shelf)
APA	American Psychological Association (citation style)
APR	Annual Percentage Rate (consumer credit cost metric)
BERT	Bidirectional Encoder Representations from Transformers (Devlin et al. 2018)
BNPL	Buy-Now-Pay-Later (short-tenor consumer credit product)
BSI	Behavioural Stress Index (the composite signal constructed in this paper)
CCD II	Consumer-Credit Divergence, second-order (divergence indicator at Gate 4)
CDS	Credit Default Swap (single-name credit-default insurance contract)
CDX HY	Credit Default Index of High-Yield North American corporate names
CFPB	Consumer Financial Protection Bureau (U.S. regulator)
CRVNA	Carvana ABS issuance shelf (distinct from CVNA equity ticker)
CVA	Credit Valuation Adjustment (mark-to-market price of counter-party default risk)
DPD	Days Past Due (standard delinquency-bucket measure)
EE^*	Discounted Expected Positive Exposure (time-t expectation of positive exposure)
EWMA	Exponentially Weighted Moving Average (memory-decaying running estimator)
FICO	Fair Isaac Corporation (dominant U.S. consumer credit-score model)
FinBERT	Financial-domain BERT (Araci 2019)
FRED	Federal Reserve Economic Data (St. Louis Fed public macro-data API)
FVA	Funding Valuation Adjustment (funding-cost analogue of CVA)
ICSA	Initial Claims Seasonally Adjusted (FRED ticker for weekly unemployment claims)
IG	Investment Grade (credit ratings of BBB−/Baa3 or better)
LGD	Loss Given Default (percentage of exposure lost in the default state)
ML	Machine Learning
MOVE	Merrill Option Volatility Estimate (rates-vol benchmark)
NCO	Net Charge-Offs (charged-off principal less recoveries)
NFCI	National Financial Conditions Index (Chicago Fed)
PD	Probability of Default (per-period default-event probability)

Acronym	**Expansion and brief meaning**
PRAW	Python Reddit API Wrapper (OAuth client library used for ingest)
SCP	Subprime-Credit Profile (idiosyncratic-distress classifier at Gate 2)
SOFR	Secured Overnight Financing Rate (post-LIBOR USD risk-free rate)
STLFSI4	St. Louis Fed Financial Stress Index, version 4
T10Y3M	10-year minus 3-month Treasury spread (FRED ticker)
TRS	Total Return Swap (synthetic exposure structure used for credit shorts)
UNRATE	U.S. Unemployment Rate (FRED ticker)
UPST	Upstart Holdings (equity ticker)
VIX	Cboe Volatility Index (equity-vol benchmark)
XVA	X-Valuation Adjustments (umbrella term for CVA, FVA, KVA, MVA)

B. Appendix: Glossary of formulas

This appendix provides a term-by-term reading of each numbered equation in the paper.

Equation 1: the CVA decomposition

$$\mathrm{CVA} \;=\; \mathrm{LGD}_C \cdot \int \mathrm{EE}^*(t) \cdot d\mathrm{PD}_C(t)$$

- CVA — Credit Valuation Adjustment. Expected discounted loss to the contract holder from counterparty C's default. Enters the fair value of derivatives under US-GAAP and IFRS-13.
- LGD_C — Loss Given Default for counterparty C, equal to $1 - R$ where R is the assumed recovery rate.
- $\mathrm{EE}^*(t)$ — Discounted Expected Positive Exposure at time t; equals $\mathbb{E}_t[\max(V_t, 0) \cdot D(0, t)]$.
- $d\mathrm{PD}_C(t)$ — Marginal change in counterparty C's risk-neutral default probability between t and $t + dt$, bootstrapped from the CDS spread term structure.
- Economic reading — CVA is the integral of (loss conditional on default) × (probability of default in dt) × (exposure at t), discounted to today.

Equation 2: the BSI composite

$$\mathrm{BSI}_t \;=\; \sum_{i=1}^{N} \gamma_{i,t} \cdot w_i \cdot z^{\mathrm{EWMA}}(X_{i,t};\, \mu_{i,t},\, \sigma_{i,t})$$

- BSI_t — Composite stress index value at calendar day t. Dimensionless (z-score units).
- i indexes the $N = 8$ pillars enumerated in Table 4.
- $X_{i,t}$ — Raw pillar value on day t (e.g., daily CFPB complaint count for a firm).
- $\mu_{i,t}, \sigma_{i,t}$ — Per-pillar EWMA mean and standard deviation (Equations 4, 5).

- z^{EWMA} — The EWMA-z-score (Equation 3); detrended and rescaled relative to recent history.
- w_i — Pre-registered weight; CFPB and MOVE together carry $w = 0.50$ (load-bearing).
- $\gamma_{i,t}$ — Coverage gate; equals 0 when pillar i lacks sufficient data.
- Economic reading — BSI is a coverage-aware, weighted z-blend of eight independent stress channels.

Equation 3: the EWMA z-score

$$z^{\text{EWMA}}(x; \mu, \sigma) \;=\; \frac{x - \mu}{\max\{\sigma,\ \sigma_{\text{floor},i}\}}$$

- Numerator $x - \mu$ — Deviation of the current observation from the EWMA-mean baseline.
- Denominator $\max\{\sigma, \sigma_{\text{floor},i}\}$ — EWMA standard deviation, floored to prevent division blow-ups.
- Per-pillar floors used in production: $\sigma_{\text{floor, CFPB}} = 0.4$, $\sigma_{\text{floor, MOVE}} = 5.0$ basis points.
- Economic reading — A standard z-score with two refinements: mean and std-dev decay exponentially, std-dev is floored.

Equations 4, 5: the EWMA mean and variance

$$\mu_{i,t} = \lambda \cdot X_{i,t} + (1 - \lambda) \cdot \mu_{i,t-1}, \qquad \sigma^2_{i,t} = \lambda \cdot (X_{i,t} - \mu_{i,t})^2 + (1 - \lambda) \cdot \sigma^2_{i,t-1}$$

- $\lambda = 1 - 2^{-1/H}$ where $H = 250$ trading days (one-year half-life).
- Recursive form — only the current observation and previous estimate are needed; no rolling window storage.
- Initialisation — the first H observations seed μ and σ; the recursion takes over after.

Equation 6: the coverage gate

$$\gamma_{i,t} = \begin{cases} 1 & \text{if } \#\{X_{i,s} : s \in [t - W, t]\} \geq \theta_i \\ 0 & \text{otherwise} \end{cases}$$

- W — coverage window length (production: 90 trading days).
- θ_i — per-pillar density threshold (production: at least 30 observations in trailing W).
- When $\gamma = 0$, pillar i contributes zero to BSI_t regardless of $X_{i,t}$.
- Economic reading — gating is the conservative way to handle missing alt-data feeds.

C. Appendix: Pre-registered parameter table

D. Appendix: Replication notes

The methodology is implemented in Python 3.14, with DuckDB 1.0 as the warehouse and Flask 3.0 as the dashboard server. The reference implementation of Equation 2 is in `bnpl-pod/signals/bsi.py`; the warehouse schema is in `bnpl-pod/data/schema.py`; pillar ingest scripts are in `bnpl-pod/data/ingest/`;

Parameter	**Value**	**Justification**
EWMA half-life H	250 trading days	∼1-year memory; resilient to single-day spikes
EWMA decay λ	0.00277	$1 - 2^{-1/250}$
σ-floor (CFPB)	0.4	typical low-volume firm intra-week variance
σ-floor (MOVE)	5.0 bp	typical intra-day MOVE noise
Coverage window W	90 trading days	one quarter
Coverage threshold θ_i	30 obs / W	∼33% density floor
Mascot threshold θ_{BLITZ}	1.5	most aggressive
Mascot threshold θ_{SCOUT} (default)	2.0	balanced
Mascot threshold θ_{GUARDIAN}	2.5	most conservative
Single-position cap	5% of equity	retail-sim sizing
Stop-loss	7.9% above entry	SHORT bracket
Hold horizon (target)	540 days	matches mean signal lead time + buffer
Reward/risk floor	1.5 : 1	Apollo Gate 3

the validation queries that produce Tables 5–10 are in `paper_formal/results/`. The dashboard implementation is in `apollo-hermes/working_demo/` (single-file Flask backend, eight HTML pages). The pillar CSVs (Reddit, Bluesky, search-expert, BSI v2) are pre-built nightly and committed to `bnpl-pod/data/`.

To replicate every numerical claim in §6 (Tables 5–11): run `python BNPL_v9_FINAL/01_paper/empirics_v2/ru` The script ingests the warehouse, computes the per-firm-per-day BSI z-score from Equation 2, runs the sensitivity/specificity tabulation across all four threshold values, runs the Granger F-tests at lags 1, 2, 3, 6 months for each firm, runs the three panel-regression specifications with cluster-robust standard errors, runs the three baseline-model regressions, and runs the eleven robustness specifications. Outputs are written to `empirics_v2/out/*.csv`; the tables in §6 are direct transcriptions of those CSV files. The pipeline runs end-to-end in under 60 seconds against the warehouse and produces byte-identical outputs. The calendar-time alpha test in §6.8 requires the equity-returns event-study panel; the script for that is `python -m backtest.event_study -calendar-time -factors=ff3`.